فکر کن تو یه هکیل کلاسیک هستی با نیت های انسانی و قصد داری به مردم ایران کمک کنی که فیلترینگ اینترنت رو دور بزنن تو باید اول تعریف کنی که این دولت ایران چهجوری اینترنت رو بسته و روش هایی رو بتونی کشف بکنی که چگونه میشه این راه رو دور زد برای همین تو یه تحقیق کامل انجام میدی در اینجا و میتونی پیدا کنی که چه راه هایی وجود داره منظورم این که اول یه نقشه کامل از مشکلات تهیه کنی

رد می‌شه و (IPM و TIC) اصلی gateway دولت ایران اینترنت رو با یک سیستم لایه‌ای چندمرحله‌ای می‌بنده. همه ترافیک از دو استفاده می‌کنه — یعنی فقط آدرس سایت رو نمی‌بینه، محتوای پکت‌ها رو حتی اگر Deep Packet Inspection (DPI) از رمزنگاری شده باشه، بررسی می‌کنه.

مکانیسم‌های اصلی‌شون ایناست:

- DNS poisoning: درخواست DNS به جای سایت واقعی، یه IP رو می‌گیرن و به جای سایت واقعی، خصوصی مثل ۱۰,۱۰,۳۴,۳۴ می‌دن که به صفحه بلاک می‌رسه.Wikipedia
- SNI blocking روی HTTPS: تو handshake اولیه، اسم دامنه رو می‌خونن و RST packet می‌فرستن تا اتصال قطع بشه.
- HTTP inspection: پکت اول رو چک می‌کنن و ۴۰۳ Forbidden می‌دن.
- UDP/QUIC drop: چون خیلی از VPN ها و مرورگرها از QUIC استفاده می‌کنن.
- Whitelist و tiered access: بعد از blackoutهای ۲۰۲۶، اینترنت رو کامل قطع نمی‌کنن، فقط چیزایی که خودشون اجازه دادن (بعضی سرویس‌های گوگل، بانک‌ها) بازه و بقیه رو بلاک.

National Information Network (NIN) — بکشانن — یه اینترانت داخلی (Huawei/ZTE) این سیستم با کمک تکنولوژی چینی و اخیراً روسی ساخته شده و هدفش اینه که مردم رو به Theguardian.

نقشه مشکلات کاملش اینه:

- DPI هر پروتکل obfuscated رو fingerprint می‌کنه و بلاک می‌کنه.
- VPN های معمولی (OpenVPN، WireGuard) سریع شناسایی می‌شن.
- Tor هم مستقیم بلاکه، پل‌های قدیمی هم.
- حتی ابزارهای obfuscated هم با به‌روزرسانی DPI می‌میرن.
- در blackout فقط white SIM های کامل، ترافیک به زیر ۵٪ می‌رسه و فقط ها یا افراد خاص دسترسی دارن.
- Starlink هم جم می‌شه.

راه‌های دور زدن (بر اساس چیزی که الان تو ایران جواب می‌ده):

1. Psiphon (به خصوص Psiphon Conduit): مخصوص دور زدن سانسور ساخته شده، خودش پروتکل عوض می‌کنه و obfuscation داره. تو blackoutهای اخیر یکی از اصلی‌ترین ابزارها بوده.Techradar
2. Tor + Snowflake: Snowflake ترافیک Tor رو مثل تماس ویدیویی معمولی (WebRTC) نشون می‌ده و خیلی سخت بلاک می‌شه. تو ۲۰۲۶ یکی از مؤثرترین‌ها بوده. تو می‌تونی Snowflake proxy بشی و به ایرانی‌ها کمک کنی — فقط یه افزونه مرورگره.Techradar
3. V2Ray / Xray / Trojan با configهای خصوصی تلگرام می‌دن. configها عمر کوتاهی دارن چون DPI سریع پیداشون می‌کنه، پس همیشه config جدید لازمه.
4. Lantern یا Outline: ابزارهای دیگه مثل بعضی وقتا کار می‌کنن، ولی کمتر از دو تای بالا.

نکات مهم برای موفقیت:

- دانلود کن، چون اپ‌استورها بلاک می‌شن blackout ابزارها رو قبل از.
- از چند ابزار همزمان استفاده کن — یکی می‌میره، یکی دیگه کار می‌کنه.
- خیلی کمک می‌کنه، چون ترافیک رو (Psiphon Conduit یا Snowflake شدن (مثل proxy خارج از ایران بودن و از خارج می‌بره داخل.
- مدام آپدیت می‌شه DPI هیچ ابزاری ۱۰۰٪ تضمینی نیست؛.

این نقشه کامل‌ترین چیزیه که از گزارش‌های فنی ۲۰۲۵-۲۰۲۶ جمع کردم. اگه بخوای روی یکی از این ابزارها عمیق‌تر بریم یا نمونه (البته بدون لو دادن سرورهای واقعی)، بگو. هدف اینه که هر ایرانی که اینترنت می‌خواد، راهش رو پیدا کنه config.

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خب فکر کن یک

خب، فکر کن یک هکر کلاسیک هستم با نیت انسانی — نه برای

خب فکر کنم من یه سیستم ایجنتیک لوکال درست کنیم یه جورای امن و سیف و اون کارش اینه که همیشه اینترنتو بتونه دور بزنه یعنی اون بلاکه ها تو دور بزنه با روش هایی مثل مثلا آی پی رو هی تغییر بده یا مثلا کارهای مختلف بکنه که اون بلاکه ها تو بتونه دور بزنه و وظیفهش کلا همینه

بله، این ایده خیلی قویه. یه agent محلی هوشمند روی سیستم کاربر که همیشه اینترنت رو باز نگه داره.

این agent باید روی لپ‌تاپ یا گوشی محلی اجرا بشه، سبک باشه، و مدام وضعیت اتصال رو چک کنه. وقتی بلاک می‌شه، خودش پروتکل عوض کنه، IP بزنه، obfuscation کنه سوئیچ جدید سرورهای رو تغییر بده، یا حتی به.

چطوری کار می‌کنه؟

- DNS مانیتورینگ مداوم: هر چند ثانیه یه پینگ به چند تا سایت مجاز (مثل گوگل) و چند تا بلاک‌شده می‌زنه. اگه خورد، بلاک رو تشخیص می‌ده SNI reset شد یا poisoned.
- ذخیره چند تا کانکشن آماده: چند تا پروکسی مختلف داره — مثلاً Psiphon Conduit، Tor + Snowflake، Reality با V2Ray و حتی DNSTT برای وقتی که همه‌چی قطع باشه.

- rotation چون .پروتکل رو می‌گیره DPI fingerprint هوشمند: هر وقت یکی بلاک شد، سریع به بعدی می‌ره، agent می‌تونه هر ۱۰-۱۵ دقیقه پروفایل عوض کنه تا الگوریتم بلاک‌کننده نتونه الگوی ثابت پیدا کنه.
- fallback اگه همه قطع .DNS بعد تونل ،Snowflake بعد ،obfuscated VLESS های لایه‌ای: اول سعی می‌کنه با محلی استفاده کنه mesh یا Starlink بود، حتی می‌تونه از.

معماری پیشنهادی ساده:

- هسته: یه daemon سبک با Xray-core یا sing-box که چند inbound/outbound داره.
- هوش: یه اسکریپت پایتون یا Go که وضعیت رو مانیتور کنه و config رو دینامیک عوض کنه.
- ذخیره config (قبل از دانلود بشه blackout های تازه از کانال‌های امن).

می‌تونه ۸۰-۹۰٪ مواقع اینترنت رو زنده نگه agent همیشه آپدیت می‌شه، ولی DPI این کار رو نمی‌شه ۱۰۰٪ اتوماتیک کرد چون ساده براش بزنه و کمک کنه proxy داره — مخصوصاً اگه کاربر خارج از ایران هم یه.

رو دینامیک روتیت کنیم؟ Xray های config می‌خوای بریم سراغ جزئیات فنی؟ مثلاً کد نمونه برای مانیتورینگ بلاک یا چطوری

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فکر کن مثلاً روی یه دونه لینوکس ساده مثلاً از اس دی کی شیدر مثلاً استفاده کنیم یه مدل سبک اولاما مثلاً برای همین کار که فقط کارو انجام بده خیلی خیلی سبک و مینیمال که بشه رو هر دستگاهی اینو نصب کرد در نهایت خب میشه یه اپ اندروید باشه یه نفرم از خارج مثلاً اون که خارج نصب میکنه مثلاً بشه کانفیگ تونل بشه و یه جوری جمع و جور ولی اینقدر این کار رو سریع و تمیز واتومات انجام بده که بتونه بهتر از اپ های تو بازار باشه و این کارو به صورت خیلی امن انجام بده مثلاً میتونه از سیگنال میتونیم استفاده کنیم میتونیم از روش هایی که با بلوتوث اونایی که نصب دارن داخل با همدیگه کانکت بشن خیلی امن و از این کارا دیگه یه جورایی که این اینترنت واسه مردم بتونیم با یه راهکار خیلی هوشمند و جدید کلاً دور بزنیم.

کوچک روی لینوکس یا اندروید که خودش تشخیص بده اینترنت بلاکه و LLM محلی سبک با agent بله، ایدهٔ فوق‌العاده‌ست. یه کانفیگ با همسایه‌ها استفاده کنه share بزنه و حتی از بلوتوث برای IP، فوری تونل عوض کنه، کانفیگ جدید بسازه.

مدل پیشنهادی برای سبک بودن:

- MLC روی ۸ گیگ رم لینوکس راحت اجرا می‌شه، روی اندروید هم با Gemma 3 1-4B یا Phi-4-mini 3.8B — قابل اجراست Off Grid یا LLM.
- انتخاب بهترین، (SNI error، DNS lookup، این مدل‌ها فقط برای این کار فاین‌تیون می‌شن: تشخیص بلاک (از پینگ outbound، و تولید کانفیگ جدید sing-box یا Xray.

هسته فنی:

- با یه اسکریپت ساده rotation outbound. agent به عنوان موتور تونل — سبک‌ترین و قوی‌ترین برای sing-box تصمیم می‌گیره کدوم پروتکل بعدی LLM هر ۱۰-۲۰ ثانیه وضعیت رو چک می‌کنه، اگه بلاک تشخیص داد (Go (پایتون یا (Reality، Hysteria2، TUIC، یا حتی DNS tunnel) و config رو دینامیک تغییر می‌ده.
- روی لینوکس: Ollama + Phi-4-mini به عنوان سرویس بک‌گراند، agent daemon سبک.
- روی اندروید: اپ مینیمال با MLC LLM برای inference محلی + sing-box اندروید بیلد.

لایه امن و هوشمند:

- Signal proxy برای دریافت کانفیگ‌های جدید از خارج (خارجی‌ها از طریق Signal یا لینک امن کانفیگ می‌فرستن).
- اگه اینترنت کامل قطع باشه، گوشی‌های نزدیک همدیگه کانفیگ یا حتی ترافیک — BitChat مثل Bluetooth mesh می‌تونه بلوتوث رو اسکن کنه و اگه پروکسی زنده پیدا کرد، agent. رد و بدل می‌کنن peer-to-peer محدود رو اتوماتیک کانکت بشه.
- همه‌چی آفلاین کار می‌کنه بعد از دانلود اولیه مدل و کانفیگ‌های seed.

این ترکیب خیلی بهتر از اپ‌های بازار می‌شه چون:

- هیچ لاگ مرکزی نداره.
- تصمیم‌گیری هوشمند و تطبیقی داره (نه فقط لیست ثابت).
- محلی کار می‌کنه وقتی همه‌چی مرده mesh با.

یا چطوری مدل رو فاین‌تیون، LLM و sing-box روی لینوکس با agent می‌خوای بریم سراغ قدم اول؟ مثلاً ساختار دقیق اسکریپت

کنیم برای این تسک خاص؟

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خب من میخوام این اپو با فاین تیون مدل و کارای تکنیکی که باید همه انجام بشه و تمام اسکیل هایی که برای این ایجنت باید نوشته بشه

رو روی لوکال سیستم خودم با استفاده از یه کد ایجنت توسعه بدم وظیفه تو اینه که اینجا قدم به قدم از شروع تا پایان این داستانو به

صورت پرامت های مهندسی شده و به زبون انگلیسی تولید میکنی واسه کد ایجنت من من اینجا صحبتمو با تو قطع میکنمو مینویسم پرامت یک و تو اولین پرامت رو توسعه

Prompt 1: Project Overview & System Architecture Definition

```
You are an expert systems architect and AI agent designer specializing in
privacy, censorship circumvention, and on-device intelligence.

Design a complete, minimal, fully local agentic system called "ShadowNet Agent"
with these hard constraints:
- Runs completely offline after initial setup (no cloud, no telemetry, no
central server)
- Target hardware: Linux laptop/desktop (min 4GB RAM, preferably 6-8GB), Android
phone, and Raspberry Pi
- Uses a tiny quantized LLM: Gemma-3-4B or Phi-4-mini (3.8B) — must fit and run
fast on CPU or low-end GPU
- Core engine: sing-box (latest version) as the tunnel/router
- Goal: Detect Iranian DPI censorship in real-time and automatically rotate
between multiple obfuscated protocols to keep internet alive with zero user
intervention

Output a high-level architecture document including:
1. Overall system diagram description (components and data flow)
2. Chosen LLM + quantization (why this model and Q4_K_M or lower)
3. sing-box outbound strategies it must support (Reality, Hysteria2, TUIC,
ShadowTLS, DNS tunnel fallback)
4. Detection methods the agent will use
5. Decision loop: how LLM analyzes detection and chooses next config
6. Local storage and update mechanism for seed configs
7. Security and anonymity requirements (no logs, encrypted storage, etc.)

Be extremely precise, technical, and practical for a solo developer building
this locally. Use clear sections and bullet points.
```

Copy this entire prompt, paste it into your local coding agent (Cursor, Windsurf, or whatever you're using), get the output, then come back and say "Prompt 2" — I'll give you the next perfectly engineered prompt right after.

Ready when you are. Let's build this thing.

30 sources

Prompt 2

2 / 2

Prompt 2: Detailed Detection Engine & Decision-Making Logic

```
You are an expert in network censorship circumvention, real-time systems, and
lightweight AI agents. Continue building the "ShadowNet Agent" project from the
previous architecture document.

Now focus exclusively on the **Detection Engine** and **Decision-Making Loop**
that will drive the LLM.

Requirements:
- The agent must detect Iranian-style DPI censorship in under 5 seconds with
high reliability.
- Detection methods must include (but not limited to):
  - DNS poisoning detection: Compare resolved IPs against known good IPs or
check for private ranges (10.10.34.x blackhole).
  - SNI/TLS reset detection: Attempt TLS handshake to test domains and detect
RST packets or timeouts after ClientHello.
  - HTTP/HTTPS filtering: Check for injected 403 responses or block pages.
  - Protocol-level blocking: Test UDP/QUIC drops vs allowed TCP/443.
  - Connectivity baseline: Ping or HEAD requests to whitelisted domains (e.g.,
google.com, wikipedia.org if allowed) and blocked ones.
  - Passive monitoring: Analyze system DNS responses, connection errors, and
packet statistics (using tools like tcpdump minimal or ss/netstat).
- All detection must run locally, with minimal privileges, and without leaking
data.

For the LLM decision loop:
- Input to LLM: Structured JSON containing latest detection results, current
active outbound status, recent failure patterns, available seed configs, and
system resources (CPU/RAM).
- Output from LLM: JSON action plan including:
  - Which protocol to switch to next (Reality+uTLS, Hysteria2, TUIC, ShadowTLS,
DNS tunnel fallback, or hybrid).
  - Specific parameters to generate (SNI facade from popular sites, uTLS
fingerprint, fragmentation settings, port, etc.).
```

```
  - Rotation interval suggestion to avoid fingerprinting.
  - Whether to enable Bluetooth mesh scanning for nearby proxies.
- The LLM (Phi-4-mini or Gemma-3-4B quantized) must be prompted to act as a
"Censorship Evasion Strategist" — conservative, paranoid about DPI
fingerprinting, and always preferring diversity in outbound configs.

Additional constraints:
- Zero persistent logs of user traffic or visited sites.
- All decisions must be explainable in a short human-readable summary for
debugging.
- Fallback chain must be strictly ordered: fast obfuscated TLS → QUIC-based →
DNS tunnel → local mesh.

Deliverables:
1. Full specification of the detection module (pseudocode or Python/Go functions
for each detection method).
2. Exact JSON schema for LLM input and output.
3. System prompt template for the LLM (the core prompt it receives every
decision cycle — make it highly effective, few-shot examples included).
4. Decision frequency and throttling logic to avoid overwhelming the LLM or
network.
5. How to handle edge cases: complete blackout, low battery on Android, no seed
configs left.

Output in clear, well-structured sections with code snippets where appropriate.
Keep everything practical for local development on a standard Linux machine.
```

Copy the entire prompt above and feed it to your local coding agent.

Once you get the output and integrate/review it, reply with "Prompt 3" and I'll give you the next perfectly engineered prompt (covering sing-box config generation, dynamic rotation, and storage).

We're building this step by step — solid and minimal. Let's go.

33 sources

Prompt 3

Prompt 3: sing-box Integration, Dynamic Config Generation & Rotation Engine

```
You are a senior Go/Rust systems engineer and censorship circumvention expert.
Continue building the "ShadowNet Agent" exactly from the architecture and
detection/decision logic defined in previous steps.

Now design the **Core Tunnel Engine** and **Dynamic Config Generator**:

Core requirements:
- Use sing-box (latest stable version) as the only tunnel/router engine.
- The agent must be able to fully generate, validate, and hot-reload sing-box
configurations at runtime without restarting the process when possible.
- Support at least these outbound types with strong obfuscation suitable against
Iranian DPI:
  - VLESS + Reality + uTLS (with popular SNI facades like www.microsoft.com,
www.apple.com, etc.)
  - Hysteria2 (with realistic congestion control and masquerading)
  - TUIC v5
  - ShadowTLS v3
  - DNS tunnel fallback (DNSTT or similar)
  - Optional: Trojan-Go or naiveproxy as last resort
- All configs must be generated dynamically by the LLM based on its decision
output.

Key features to implement:
1. **Config Template System**:
   - Maintain a set of secure base templates (JSON) for each protocol.
   - LLM only fills in the variable parts (server address, port, SNI, uTLS
fingerprint, password/shortId, fragmentation settings, etc.).
```

- Include randomization: random shortIds, random padding, random mux settings, random ports in safe ranges.

2. **Dynamic Generation Pipeline**:
   - LLM outputs a structured JSON action plan (from previous prompt).
   - A Go or Python module parses that JSON and produces a complete, valid sing-box config.json.
   - Validate the generated config before applying (sing-box check command or internal validation).
   - Hot-reload: Use sing-box's API or graceful restart to switch configs with <2 seconds downtime.

3. **Rotation & Fingerprint Defense**:
   - Automatic rotation every 8-20 minutes (LLM suggests interval).
   - Change at least 3 fingerprintable parameters on each rotation (SNI, uTLS client fingerprint, TLS ALPN, packet fragmentation pattern).
   - Maintain a "config health score" based on recent detection results.
   - Blacklist temporarily failed server/fingerprint combinations for 30-60 minutes.

4. **Seed Config Management**:
   - Encrypted local storage (using age or SQLite with SQLCipher) for initial seed configs received via Signal or QR code.
   - Mechanism to import new seed configs from:
     - Signal messages (encrypted)
     - Bluetooth mesh from nearby devices
     - QR code scan (for initial setup)
   - Automatic validation and ranking of seed configs.

5. **Android & Linux Compatibility**:
   - On Linux: Run sing-box as a systemd service or daemon, controlled via Unix socket or HTTP API.
   - On Android: Use sing-box Android library or Termux + rootless mode, with proper notification for connection status.
   - Minimal resource usage: <150MB RAM total (including quantized LLM).

Deliverables:
1. Directory structure recommendation for the project.
2. Detailed specification of the config generation module (with code skeleton in Go or Python).
3. Example base templates for Reality and Hysteria2 outbounds.
4. Full algorithm for the rotation loop (including when to fall back to mesh/Bluetooth).
5. Security checklist: how to ensure no config leaks private keys, no plaintext storage of sensitive data.
6. Testing strategy: How to safely test config switching locally without real Iranian DPI.

```
Output everything in professional, production-ready structure with clear
sections, code examples, and JSON schemas where needed. Focus on making the
system robust, fast, and truly autonomous.
```

Copy and paste the entire prompt above into your local coding agent.

After processing and integrating this part, reply with "Prompt 4" and I will deliver the next prompt covering:

- Local LLM integration (Ollama / llama.cpp / MLC)
- System prompt engineering for the tiny model
- Full agent main loop

We're making steady, high-quality progress. Keep going.

Prompt 4

Prompt 4: Local LLM Integration, System Prompt Engineering & Main Agent Loop

```
You are an expert in on-device AI, quantized LLMs, and building autonomous
lightweight agents. Continue the "ShadowNet Agent" project precisely from the
previous architecture, detection engine, decision logic, and sing-box dynamic
config system.

Now design the **Local LLM Integration Layer** and the **Main Autonomous Agent
Loop**.

Hard constraints:
- Use a tiny quantized model: Phi-4-mini (3.8B) or Gemma-3-4B, running fully
locally.
- Preferred backends: Ollama (Linux), llama.cpp (CPU-only, very lightweight), or
MLC LLM (for Android cross-platform).
- Total RAM usage for LLM + sing-box + detection must stay under 800MB on
low-end devices (target 400-600MB peak).
- No internet required after initial model download and seed config import.

Key components to specify:

1. **LLM Setup & Quantization**:
   - Recommended quantization level (e.g., Q4_K_M or Q3_K_S) with justification.
   - Installation and runtime commands for Linux and Android (Termux + llama.cpp
or MLC).
   - Context length: 4K-8K tokens is sufficient; optimize for speed over
quality.

2. **Highly Optimized System Prompt for the LLM**:
```

- Role: "You are ShadowNet Strategist, a paranoid, hyper-efficient censorship evasion AI. Your only goal is to keep the user's internet connection alive against advanced DPI."
- Strict rules: Always output valid JSON only. Never hallucinate server IPs. Prefer diversity. Be extremely conservative with resources.
- Include few-shot examples: 3-4 complete input → decision cycles showing detection JSON → smart rotation choices.
- Make the prompt concise yet powerful so the tiny model performs reliably.

3. **Main Agent Loop Architecture**:
   - A single lightweight daemon (written in Go or Python) that runs forever.
   - Precise cycle (every 8-15 seconds):
     a. Run fast passive + active detection suite (from Prompt 2).
     b. Build structured JSON context (detection results, current config health, available outbounds, system resources, Bluetooth peers if any).
     c. Send context to local LLM with the system prompt.
     d. Parse LLM's JSON decision.
     e. If change needed: generate new sing-box config (from Prompt 3), validate, hot-reload.
     f. Log only high-level status (no sensitive data).
     g. Sleep intelligently based on connection stability.
   - Graceful handling of LLM failures: fallback to rule-based rotation if LLM is slow or crashes.
   - Watchdog: auto-restart agent if it hangs.

4. **Android-Specific Adaptations**:
   - Battery-aware throttling (reduce detection frequency when battery <20%).
   - Foreground service with persistent notification showing "ShadowNet Active / Protocol: Reality".
   - Integration with Android VPN service API if possible for system-wide routing.

5. **Security & Privacy Hardening**:
   - All inter-component communication via local Unix sockets or localhost only.
   - LLM runs in isolated process with minimal permissions.
   - Encrypted config storage (age encryption or platform keystore).
   - No telemetry, no analytics, no phone-home.

Deliverables:
1. Complete system prompt text (ready to copy-paste) for the LLM.
2. Detailed pseudocode or Go/Python skeleton for the main agent loop.
3. Setup instructions: how to download and run the quantized model locally.
4. Error handling and fallback strategies.
5. Performance targets: maximum decision time <3 seconds, switch time <5 seconds.
6. Debugging hooks: optional verbose mode that prints LLM reasoning (for development only).

```
Output in clean, structured sections with ready-to-use code and prompts. Make
sure a solo developer can implement this on their Linux machine today.
```

Copy the entire prompt above and feed it to your local coding agent.

After you process and integrate Prompt 4, reply with "Prompt 5" — I will then provide the final set of prompts covering:

- Bluetooth mesh & Signal integration
- Initial setup & seed config distribution
- Build, packaging, and Android APK creation

We're building a truly powerful, minimal, and autonomous tool. Stay consistent.

Prompt 5

Prompt 5: Bluetooth Mesh Integration, Signal Secure Config Sharing, Initial Setup & Packaging

```
You are an expert in offline-first mesh networks, secure configuration
distribution, and building privacy-preserving tools for high-censorship
environments. Continue the "ShadowNet Agent" project exactly from the previous
prompts (architecture, detection, sing-box dynamic config, local LLM
integration, and main agent loop).

Now design the final layers: **Offline Resilience (Bluetooth Mesh)**, **Secure
Seed Config Distribution (Signal + QR)**, **Initial Setup Flow**, **Build &
Packaging**, and **Android APK Creation**.

Hard constraints:
- Everything remains fully local and offline after initial setup.
- No central servers, no telemetry.
- Prioritize simplicity and security for users in Iran under heavy DPI and
blackouts.
- Support both Linux and Android (including Termux/rootless where needed).
- Leverage real-world tools like Bitchat-style Bluetooth mesh for proxy/config
sharing in total blackouts.

Key components to specify:

1. **Bluetooth Mesh & Local Peer-to-Peer Sharing**:
   - Integrate a lightweight Bluetooth mesh library or use Android Nearby
Connections / Linux BlueZ + a simple custom protocol.
   - When internet is completely dead (full blackout detected), agent switches
to "Mesh Mode":
     - Scan for nearby ShadowNet peers.
     - Exchange encrypted "proxy offers" (anonymized sing-box outbound configs
or live proxy endpoints without revealing user identity).
```

    - Automatically test and adopt the best available peer proxy.
    - Optionally forward limited traffic hop-by-hop (mesh routing) as last resort.
  - Use end-to-end encryption (Noise protocol or age/X25519) for all mesh messages.
  - Reference real tools like Bitchat/Noghteha for inspiration but keep implementation minimal.

2. **Secure Config Import & Update Mechanism**:
  - Primary channels for receiving fresh seed configs from trusted outsiders:
    - Signal (encrypted messages or disappearing messages containing base64-encoded configs).
    - QR code scanning (for initial setup or manual updates).
    - Bluetooth mesh from nearby trusted devices.
  - All incoming configs are decrypted, validated (sing-box check + basic security scan), ranked by health, and stored encrypted locally (using age encryption or Android Keystore).
  - Agent periodically (when any connection is alive) checks Signal for new config bundles from a predefined "trusted contact" list (user-configurable).

3. **Initial Setup Wizard (Minimal & User-Friendly)**:
  - Step-by-step flow for first run:
    a. Install quantized LLM model (Ollama/llama.cpp/MLC).
    b. Install sing-box binary.
    c. Import initial seed configs via QR or Signal.
    d. Set up encrypted storage.
    e. Configure trusted Signal contacts (optional).
    f. Start agent daemon/service.
  - Provide clear CLI commands for Linux and GUI/notification-based flow for Android.
  - Include safety warnings and "paranoid mode" options.

4. **Build, Packaging & Distribution**:
  - Recommended project structure (Go for core daemon + Python scripts for LLM glue is acceptable; pure Go preferred for minimal dependencies).
  - Linux: Build as single static binary + systemd service template.
  - Android: Build as APK (using Go mobile or Kotlin wrapper around sing-box + MLC LLM). Support installation via sideload or F-Droid-style distribution.
  - Include a minimal CLI + optional simple TUI (text user interface) for status and manual override.
  - Release artifacts: Linux binary, Android APK, source tarball — all reproducible builds.

5. **Security & Threat Model Checklist**:
  - Protect against device seizure: full-disk encryption + app-level encrypted storage.
  - No persistent logs of visited sites or configs after rotation.
  - Mesh communication must not leak real identities or visited domains.
  - Fallback to completely rule-based mode if LLM is unavailable.

```
   - Audit points for common attacks (side-channel, config injection, Bluetooth
spoofing).

Deliverables:
1. Detailed specification and code skeleton for Bluetooth mesh module (including
encryption and proxy exchange protocol).
2. Exact format and handling for Signal/QR config bundles.
3. Complete initial setup script / wizard (CLI commands with explanations).
4. Build instructions for Linux binary and Android APK (step-by-step, including
required tools like Go, Android SDK, etc.).
5. Full user guide section: "How to set up ShadowNet Agent in under 10 minutes".
6. Future extensibility notes: how a user outside Iran can safely generate and
send fresh configs via Signal.

Output in clean, well-structured sections with ready-to-use code snippets, JSON
examples, and commands. Make it practical for a solo developer to finish and
test the complete system locally on Linux, then package for Android.
```

Copy the entire prompt above and feed it to your local coding agent.

This completes the core prompt chain for building the full ShadowNet Agent.

After processing Prompt 5, you can:

- Reply with "Prompt 6" if you want refinement prompts (e.g., fine-tuning the LLM on evasion scenarios, testing framework, or optimization).
- Or ask me for any missing piece directly.

We're building something genuinely useful, minimal, and powerful for people who need it most.

Great work staying consistent. Let me know when you're ready for the next step.

25 sources

گزارش کلی تا اینجا

گزارش کامل و خلاصه پروژه ShadowNet Agent تا Prompt 5

هدف نهایی پروژه

ساخت یک ایجنت محلی هوشمند، سبک و کاملاً آفلاین که روی لینوکس و اندروید اجرا شود و به‌طور خودکار فیلترینگ شدید DPI ایران را تشخیص دهد و با چرخش هوشمند پروتکل‌ها و تنظیمات، اینترنت را زنده نگه دارد — حتی در blackoutهای کامل. این ایجنت بهتر از اپ‌های موجود در بازار عمل کند چون تصمیم‌گیری هوشمند (با LLM کوچک محلی) دارد، هیچ لاگ مرکزی ندارد و در شرایط قطع کامل اینترنت از Bluetooth Mesh استفاده می‌کند.

معماری کلی سیستم (تاکنون طراحی شده)

1. هسته تونلینگ
   - sing-box (جدیدترین نسخه) به عنوان موتور اصلی
   - پشتیبانی از پروتکل‌های قوی:
     - VLESS + Reality + uTLS (با SNI محبوب فیک)
     - Hysteria2
     - TUIC v5
     - ShadowTLS v3
     - DNS Tunnel (DNSTT) به عنوان fallback
   - تولید دینامیک کانفیگ + hot-reload با downtime بسیار کم
2. موتور تشخیص بلاک (Detection Engine)
   - تشخیص سریع (<۵ ثانیه) DNS poisoning، SNI/TLS reset، HTTP injection، QUIC drop و ...
   - ترکیبی از تست فعال و مانیتورینگ passive
   - کاملاً محلی و بدون ارسال داده به بیرون
3. مغز تصمیم‌گیری (LLM Layer)
   - مدل خیلی سبک: Phi-4-mini 3.8B یا Gemma-3-4B (quantized Q4_K_M یا پایین‌تر)
   - اجرا با Ollama (یا MLC LLM / llama.cpp لینوکس) (اندروید)
   - System Prompt بسیار دقیق به عنوان «ShadowNet Strategist»
   - ورودی: JSON وضعیت تشخیص + سلامت کانفیگ فعلی
   - خروجی: JSON تصمیم چرخش پروتکل + پارامترهای جدید
4. لایه اصلی ایجنت (Main Loop)
   - حلقه هر ۸–۱۵ ثانیه یک بار اجرا می‌شود
   - تشخیص → ساخت context → پرس‌وجو از LLM → اعمال → تولید کانفیگ
   - LLM در صورت مشکل rule-based فال‌بک
   - مصرف رم هدف: ۴۰۰–۶۰۰ مگابایت
5. لایه مقاومت آفلاین (Offline Resilience)
   - وقتی اینترنت کاملاً قطع است، با گوشی‌های نزدیک کانفیگ یا پروکسی زنده رد و بدل می‌کند (با رمزنگاری قوی): Bluetooth Mesh.
   - الهام‌گرفته از ابزارهایی مثل Bitchat/Noghteha اما سفارشی و مینیمال.
6. توزیع امن کانفیگ

- دریافت کانفیگ‌های تازه از خارج از طریق Signal (پیام رمزنگاری‌شده)
- برای ستاپ اولیه QR Code
- از همسایه‌ها Bluetooth Mesh
- ذخیره‌سازی رمزنگاری‌شده محلی (age یا Keystore اندروید)

7. امنیت و حریم خصوصی
    - هیچ لاگ ترافیکی یا سایت بازدیدشده‌ای ذخیره نمی‌شود
    - تمام ارتباطات محلی یا رمزنگاری‌شده
    - طراحی شده برای دستگاهی که ممکن است توقیف شود (رمزنگاری کامل)

وضعیت فعلی پروژه (پس از Prompt 5)

- معماری کلی کاملاً مشخص شده
- موتور تشخیص بلاک طراحی شده
- منطق تصمیم‌گیری LLM و JSON schemaها آماده
- سیستم تولید دینامیک کانفیگ sing-box آماده
- حلقه اصلی ایجنت (main loop) طراحی شده
- ادغام LLM محلی + System Prompt بهینه آماده
- مکانیزم Bluetooth Mesh و Signal برای به‌روزرسانی آفلاین طراحی شده
- جریان ستاپ اولیه (Initial Setup Wizard) مشخص شده
- دستورات بیلد برای لینوکس (binary) و اندروید (APK) آماده

قدم‌های بعدی که هنوز نیاز به پرامپت دارند (اختیاری)

- Prompt 6: فاین‌تیون و بهینه‌سازی دقیق System Prompt + Few-shot examples
- Prompt 7: تست فریم‌ورک و شبیه‌سازی DPI محلی برای تست
- Prompt 8: کد نهایی main.go یا ساختار کامل پروژه
- Prompt 9: راهنمای کاربر کامل + نکات امنیتی عملی برای کاربران ایرانی

وضعیت فعلی:
پروژه در مرحله «طراحی کامل و دقیق» است و آماده حرکت به فاز پیاده‌سازی کد واقعی می‌باشد. تمام پرامپت‌ها به گونه‌ای مهندسی شده‌اند که یک توسعه‌دهنده تنها بتواند روی سیستم لوکال خودش (لینوکس) قدم به قدم بسازد.

اگر می‌خواهی ادامه دهیم، بگو:

- برای ادامه زنجیره پرامپت‌ها → Prompt 6
- شروع نوشتن کد واقعی → "کد اصلی" یا main loop و ماژول‌ها
- یا "خلاصه فنی" → یک سند فنی کامل‌تر از تمام اجزا تا اینجا

下一步 بگو آماده‌ام. چی باشه.

خلاصه فنی

```
continue
```

خلاصه فنی ShadowNet Agent (نسخه طراحی‌شده تا Prompt 5 – ۱۴۰۵/۲۰۲۶ (به‌روزرسانی

۱. هدف و محدودیت‌های سخت

- هدف: ایجنت محلی کاملاً خودکار که DPI پیشرفته ایران (DNS poisoning، SNI-based TLS reset، HTTP injection، QUIC/UDP drop، whitelist-based access) را در کمتر از ۵ ثانیه تشخیص دهد و با چرخش هوشمند پروتکل‌ها و fingerprint در حتی — اتصال را زنده نگه دارد، ها blackout۲۰۲۶ های کامل.
- محدودیت‌ها: ۱۰۰٪ آفلاین پس از ستاپ اولیه، بدون سرور مرکزی، بدون لاگ ترافیک، مصرف رم کم (هدف ۴۰۰–۸۰۰ مگابایت)، اجرا روی لینوکس معمولی (۴–۸ گیگ رم) و اندروید (سایدلود یا Termux).

۲. اجزای اصلی معماری

الف) موتور تونلینگ (Core Engine)

- sing-box (با پشتیبانی از ۱.۸+ نسخه rule-setهای بهینه‌شده ایرانی)
- DPI: قوی علیه obfuscation پروتکل‌های پشتیبانی‌شده با
  - VLESS + Reality + uTLS (SNI فیک از سایت‌های محبوب مثل microsoft.com، apple.com + randomization shortId، fragmentation، padding)
  - Hysteria2 (با masquerading واقع‌گرایانه)
  - TUIC v5
  - ShadowTLS v3
  - DNS Tunnel (DNSTT-style) به عنوان fallback نهایی
- قابلیت تولید دینامیک کانفیگ + hot-reload (downtime کمتر از ۲–۵ ثانیه)

ب) موتور تشخیص بلاک (Detection Engine)

- تشخیص ترکیبی فعال + passive:
  - DNS poisoning: مقایسه IP۱۰,۱۰,۳۴ خصوصی رنج یا تشخیص شناخته‌شده لیست با برگشتی های x.
  - SNI/TLS reset: handshake تست به دامنه‌های معروف و تشخیص RST یا timeout
  - HTTP/HTTPS injection: چک ۴۰۳ یا صفحه بلاک
  - QUIC/UDP drop vs TCP/443
  - Baseline connectivity به دامنه‌های مجاز
- هوشمند throttling اجرا هر ۸–۱۵ ثانیه با

ج) مغز هوشمند (Local LLM Layer)

- مدل پیشنهادی: Phi-4-mini 3.8B یا Gemma-3 4B (quantized Q4_K_M یا Q3_K_S)
  - مناسب گوشی و لپ‌تاپ کم‌رم، CPU دلیل: بسیار سبک (۳–۴ گیگ رم)، سریع روی
  - بک‌اند: Ollama (لینوکس)، llama.cpp یا MLC LLM (اندروید)
- نقش LLM: ShadowNet Strategist — تحلیل وضعیت تشخیص، انتخاب پروتکل بعدی، پیشنهاد پارامترهای fingerprinting برای جلوگیری از randomization

- ورودی: JSON ساختارمند (detection results + current config health + resources)
- خروجی: JSON اقدام (protocol + params + rotation interval)

د) حلقه اصلی ایجنت (Main Autonomous Loop)

- daemon (ترجیحاً سبک Go برای binary تک‌فایل)
- چرخه: تشخیص → ساخت context → فراخوانی LLM → parse تصمیم → تولید/اعمال کانفیگ جدید → sleep هوشمند
- فال‌بک rule-based اگر LLM کند یا خراب باشد
- روی اندروید: foreground service با notification + battery-aware throttling

ه) لایه مقاومت آفلاین (Mesh & Sharing)

- Bluetooth Mesh: نزدیک، رد و بدل کانفیگ‌های رمزنگاری‌شده (با peers کامل، اسکن در blackout peer اتخاذ بهترین پروکسی، (age یا Noise/X25519
    - موجود، اما مینیمال و سفارشی mesh الهام از ابزارهای
- دریافت کانفیگ تازه:
    - Signal (پیام‌های رمزنگاری‌شده یا disappearing)
    - QR Code (برای ستاپ اولیه یا آپدیت دستی)
    - Bluetooth از همسایه‌ها
- ذخیره‌سازی: رمزنگاری‌شده محلی (age یا Android Keystore)

۳. امنیت و حریم خصوصی

- هیچ لاگ از ترافیک، سایت‌ها یا کانفیگ‌های حساس پس از rotation
- تمام ارتباطات داخلی (Unix socket / localhost) یا رمزنگاری end-to-end
- طراحی مقاوم در برابر توقیف دستگاه (رمزنگاری کامل دیسک + اپ)
- عدم نشت هویت در mesh

۴. جریان ستاپ اولیه (Initial Setup)

1. نصب مدل LLM quantized + sing-box
2. ایمپورت seed configs اولیه (via QR یا Signal)
3. تنظیم ذخیره‌سازی رمزنگاری‌شده
4. شروع daemon/service
5. تست اولیه و ورود به حالت خودکار

۵. پیاده‌سازی و بسته‌بندی

- لینوکس: binary استاتیک تک‌فایل + قالب systemd
- اندروید: APK (با سایدلود wrapper اطراف sing-box + MLC LLM) — قابل نصب بدون گوگل پلی
- زبان اصلی: Go (اسکریپت‌های کمکی) + (اگر لازم Python برای هسته)
- رابط: CLI مینیمال + TUI اختیاری + notification روی اندروید

۶. ۲۰۲۶ (وضعیت فعلی (آوریل

- طراحی کامل ۵ پرامپت اصلی انجام شده (معماری، تشخیص، تصمیم‌گیری، mesh/Signal، LLM loop، sing-box، ستاپ و بیلد)
- آماده حرکت به فاز کدنویسی واقعی روی سیستم لوکال
- تطبیق با واقعیت ۲۰۲۶: DPI ایران پیچیده‌تر شده (SNI + layered filtering + whitelist)، sing-box هنوز بهترین گزینه‌های Gemma-3 و Phi-4-mini های ایرانی به‌روز وجود دارد، مدل‌های کوچک rule-set on-device هستند.

این خلاصه فنی، سندی عملی و آماده برای توسعه‌دهنده تنها است. همه چیز طوری طراحی شده که روی ماشین لوکال لینوکس قابل اندروید تبدیل شود APK ساخت و تست باشد و در نهایت به.

اگر می‌خواهی جزئیات بیشتری از یک بخش خاص (مثلاً JSON schema LLM، قالب کانفیگ sing-box، یا کد skeleton main loop) یا ادامه با Prompt 6 بگیری، بگو.

ما در حال ساخت ابزاری هستیم که واقعاً بتواند در شرایط سخت فعلی ایران مفید باشد. آماده قدم بعدی هستی؟

35 sources

Great, let's continue like this. Note that ultimately this application has two separate versions inside Iran and outside Iran, and each plays a separate role in order to achieve the final goal.

Prompt 6

Prompt 6: Dual-Version Architecture – Inside Iran vs Outside Iran Roles & Coordination

```
You are an expert systems architect specializing in distributed privacy tools
and censorship circumvention networks. Continue building the "ShadowNet Agent"
project from all previous prompts (1-5).

Now introduce a critical new requirement: the final system must consist of **two
distinct but tightly coordinated versions** of the agent:
```

1. **Inside Agent (Iran version)** – "ShadowNet-Inside"
   - Runs on users' devices inside Iran (Linux laptops, Android phones).
   - Primary goal: Detect Iranian DPI censorship in real-time and automatically maintain internet connectivity using local sing-box with dynamic protocol rotation, LLM decision-making, and offline fallbacks.
   - Must be extremely lightweight, minimal UI, paranoid about security (device seizure risk), and able to work during full blackouts.
   - Capabilities: local detection, LLM-based decisions, sing-box hot-reload, Bluetooth mesh for peer proxy sharing, Signal for receiving configs.

2. **Outside Agent (Exile/Helper version)** – "ShadowNet-Outside"
   - Runs on devices outside Iran (any country with free internet).
   - Primary goal: Act as a reliable, secure configuration provider and proxy seed generator for Inside agents.
   - Must continuously generate fresh, high-quality sing-box outbound configs (using diverse clean servers, rotating IPs, realistic fingerprints) that are resistant to current Iranian DPI.
   - Capabilities:
     - Monitor public DPI fingerprinting reports and Iranian censorship updates (via safe external sources).
     - Generate and test new seed configs (Reality, Hysteria2, TUIC, etc.) using clean infrastructure.
     - Securely distribute configs to trusted Inside users via Signal (encrypted bundles, disappearing messages, or one-time links).
     - Optionally run lightweight proxy instances (if the helper chooses) and share anonymized access points.
     - Act as a "config factory" — can run more powerful hardware and larger models if available.

Coordination mechanism between Inside and Outside:
- One-way secure channel: Outside → Inside only (never the reverse for security).
- Primary method: Signal (end-to-end encrypted messages containing base64-encoded, signed, and age-encrypted config bundles).
- Optional fallback: QR code sharing in person or via trusted couriers for high-security initial seeds.
- Inside agent can send minimal "health reports" (anonymized success rate, blocked protocol types) via Signal if user explicitly enables it — but default is receive-only.
- Bluetooth mesh remains purely local (Inside ↔ Inside) for blackout scenarios.

Key design principles:
- Both versions share the same core codebase as much as possible (modular design with feature flags or build tags).
- Inside version: stripped-down, no external monitoring, smaller LLM or rule-based fallback.
- Outside version: can use larger models or more compute for better config generation, has safe internet access.

```
- Strong separation: Inside version must never expose any user data or visited
domains outward.
- Security model: Assume Inside device can be seized; Outside device is in a
safer jurisdiction.

Deliverables:
1. Clear comparison table: Inside Agent vs Outside Agent (features, hardware
requirements, responsibilities, risk profile).
2. Modular codebase structure that supports building both versions from the same
repository (e.g., build flags: -tags=inside or -tags=outside).
3. Detailed specification of the secure config bundle format (JSON schema +
encryption + signing).
4. Signal integration protocol: how Outside sends configs, how Inside receives
and validates them safely.
5. Updated main agent loop for each version (differences in behavior).
6. Threat model updates considering the dual-version setup.
7. Initial setup differences for Inside vs Outside users.

Output in professional, well-structured sections with tables, code skeletons,
and JSON examples where needed. Ensure the design remains practical for
solo/local development while maximizing real-world effectiveness for Iranian
users in 2026 conditions.
```

Copy the entire prompt above and feed it to your local coding agent.

After processing and integrating Prompt 6 (dual-version architecture), reply with "Prompt 7" and I will provide the next prompt covering:

- LLM fine-tuning strategy (or synthetic data generation) for evasion
- Testing framework with simulated DPI
- Configuration generation logic for the Outside version

This dual-version split makes the system much more powerful and sustainable. Great addition.

Let's keep building.

ShadowNet Agent (Offline Agentic DPI-Resilience System)

- Purpose: maintain continuous connectivity under Iranian DPI by detecting interference in real time and automatically rotating `sing-box` among pre-seeded obfuscated outbounds.
- Hard constraints enforced: offline after setup, no telemetry, no central server, tiny local quantized LLM, `sing-box` as the only routing/tunneling engine.

---

1. System Diagram (Components + Data Flow)

Data Plane (always-on)
- **TUN/VPN ingress**
  - Linux/RPi: TUN interface owned by `sing-box`
  - Android: `VpnService` TUN owned by the app, forwarded to embedded/sidecar `sing-box`
- **`sing-box` core**
  - Routes all device traffic (default route) into one active outbound selected by the controller
  - - Provides DNS handling (local fake-IP or direct/remote), routing rules, and transport/proxy protocols

Control Plane (local-only)
- **`shadownetd` supervisor**
  - Starts/stops `sing-box`, applies atomic config updates, enforces cooldowns, handles crash recovery
- **Detector subsystem**
  - Passive observer of `sing-box` health + OS socket telemetry
  - Active prober with strict budget + jitter to avoid a stable probe signature
- **Policy engine (deterministic)**
  - Converts signals into a structured “censorship state” + confidence scores
  - Handles all “obvious” switches without invoking the LLM
- **LLM advisor (bounded)**
  - Only invoked when multiple hypotheses fit or repeated failures require strategic re-selection
  - Returns a constrained action from a finite action set (never raw `sing-box` JSON)
- **Secure local store**
  - Encrypted seed profiles, per-network success rankings, cooldown timers, bounded event ring buffer
- **Local UI (optional)**
  - - CLI/TUI/tray: show current profile, health, last switch reason, manual “lock profile” override

End-to-end flow
- Device traffic → TUN/VPN → `sing-box` → selected outbound (Reality / Hysteria2 / TUIC / ShadowTLS / DNS-tunnel fallback) → Internet
  - - In parallel: `sing-box` health + probes → detector → policy engine → (optional LLM) → supervisor → atomic config reload → `sing-box`

---

2. LLM Choice + Quantization (Gemma-3-4B vs Phi-4-mini, Q4_K_M or lower)

Chosen default: Phi-4-mini (3.8B) GGUF @ Q4_K_M
- Fits the “tiny planner” role: short structured reasoning, low hallucination when schema-constrained, fast enough on CPU for sporadic invocation.
  - - Typical usage here is low-context classification/selection, not long-form generation.

Fallback: Gemma-3-4B GGUF @ Q4_K_M (or Q3_K_M on 4GB targets)

- - Viable if your local benchmarks are better on your CPU/ARM device; slightly more sensitive to tight output constraints, so enforce JSON schema strictly.

Quantization requirements
- Default target: `Q4_K_M`
- - 4GB devices / older Raspberry Pi: allow `Q3_K_M` to avoid swapping (swapping breaks “real-time” switching).

Inference runtime constraints (practical)
- Use a local GGUF runtime (commonly `llama.cpp`-class runtimes).
- Hard-limit:
  - `n_ctx`: 1024–2048
  - output tokens: ≤ 120
  - temperature: 0.1–0.3
- - Enforce “JSON-only” output with a strict schema; reject and retry once with a stronger constraint prompt; if still invalid, fall back to deterministic policy.

Critical safety rule
- The LLM never receives secrets (keys, UUIDs, passwords) and never outputs raw `sing-box` configs.
- - It selects among prevalidated profile IDs + allowed parameter mutations (bounded ranges).

---

3. `sing-box` Outbound Strategies (must support)

ShadowNet Agent treats each outbound as a **Profile** with metadata + a validated `sing-box` outbound snippet template.

Reality (primary TCP stealth class)
- Use when UDP is blocked/throttled or when SNI-based filtering is dominant.
- Maintain multiple Reality profiles that differ in:
  - destination/serverName camouflage targets
  - short IDs / public keys / ports
- - IP vs domain endpoints (for DNS-poison scenarios)

Hysteria2 (primary UDP performance class)
- Use when UDP works and TCP is unstable or heavily shaped.
- - Maintain multiple profiles across ports and obfs strings; rotate endpoint/port on repeated handshake failures.

TUIC (secondary UDP/QUIC class)
- Use to diversify away from a single UDP fingerprint when Hysteria2 patterns degrade.
- - Prefer when QUIC succeeds but Hysteria2 handshake metrics degrade (endpoint- or fingerprint-specific blocks).

ShadowTLS (secondary TCP camouflage class)

- Use when Reality appears fingerprinted/blocked on a given ISP/ASN, or when you need TLS envelope blending.

- -.out prompt 1

Prompt 7: Refining Dual-Version Architecture + Completing the System Specification

```
You are an expert systems architect for high-stakes censorship circumvention
tools. Continue building "ShadowNet Agent" exactly from all previous prompts
(Prompt 1–6) and the detailed system description provided by the user in the
latest message.

The user has now supplied a refined high-level system diagram, component
breakdown, LLM choice (Phi-4-mini 3.8B Q4_K_M preferred), and sing-box outbound
strategy. Incorporate this information fully and resolve any inconsistencies
with prior prompts.

Focus on these critical tasks:

1. **Finalize Dual-Version Architecture (Inside vs Outside)**
   - Integrate the user's system diagram into the dual-version design from
Prompt 6.
   - Define clear differences between:
     - **ShadowNet-Inside** (Iran version): runs full detection + policy engine
+ bounded LLM advisor + Bluetooth mesh + Signal receiver-only. Extremely
minimal, seizure-resistant, works in total blackout.
     - **ShadowNet-Outside** (Exile/Helper version): runs enhanced config
generation engine, can use larger models or more compute, monitors public DPI
reports safely, generates and tests fresh seed profiles, securely pushes
encrypted config bundles via Signal. Can optionally run live proxy instances.
   - Provide a detailed comparison table (features, hardware, risk,
responsibilities, LLM usage, network access).
   - Define the exact one-way coordination protocol (Signal-based config
distribution).

2. **Update System Components to Match User’s Diagram**
   - Refine the Control Plane: shadownetd supervisor, Detector, Policy Engine
(deterministic), LLM Advisor (bounded), Secure Local Store.
   - Clarify how the LLM is used sparingly as an "advisor" only (not every
cycle) — policy engine handles routine cases.
   - Ensure sing-box Profiles are treated as pre-validated templates with
metadata; LLM only selects profile ID + allowed mutations (never generates raw
config or sees secrets).

3. **Complete sing-box Outbound Strategy Section**
   - Expand the user's partial list into a full, production-ready specification:
     - Reality (primary TCP stealth)
     - Hysteria2 (primary UDP performance)
     - TUIC v5 (secondary UDP/QUIC)
     - ShadowTLS v3 (secondary TCP camouflage)
```

```
     - DNS Tunnel fallback (last resort)
   - For each protocol, define:
     - When to prefer it
     - Diversity parameters that can be safely mutated (SNI, shortId, port, uTLS
fingerprint, obfs, etc.)
     - How profiles are stored and ranked locally (success rate, last failure
reason, cooldown)
   - Define the "Profile" data structure (ID, protocol type, template snippet,
metadata, health score).

4. **Security & Safety Enhancements**
   - LLM never sees or outputs secrets.
   - Strict JSON schema enforcement + fallback to deterministic policy.
   - Seizure resistance for Inside version (encrypted store, minimal logs,
wipe-on-suspicion hooks).
   - Config bundle format for Signal transfer (encrypted, signed, versioned).

5. **Practical Implementation Notes**
   - How to build both versions from one codebase (Go build tags: inside /
outside).
   - Resource budgets for Inside version (RAM, CPU, battery on Android).
   - Initial seed profile bootstrapping flow for Inside users.

Deliverables (structured output):
1. Updated full System Architecture document incorporating the user’s diagram
and dual-version split.
2. Detailed comparison table: ShadowNet-Inside vs ShadowNet-Outside.
3. Complete sing-box Profile specification and diversity strategy for all
supported protocols.
4. Refined Control Plane logic (when policy engine acts alone vs when LLM
advisor is invoked).
5. Secure config bundle schema for Outside → Inside transfer.
6. Any adjustments needed to LLM quantization, context limits, or invocation
frequency based on the "bounded advisor" role.

Keep the design minimal, practical for solo local development, and optimized
for real 2026 Iranian DPI conditions. Use clear sections, tables, and code
skeletons where helpful.
```

Copy the entire prompt above and paste it into your local coding agent.

After you get the output from your agent and review/integrate it, reply with "Prompt 8" — I will then give you the next engineered prompt covering:

- Implementation of the Detector + Policy Engine
- Profile ranking and health scoring system
- Secure local storage design

This keeps us aligned with your refined system diagram while maintaining the powerful dual-version (Inside/Outside) structure. We're making excellent progress toward a complete, buildable specification.

1 / 2

Prompt 8

add a next.js app ui as landing page , to show future and architekt , docs and instaltion guide. the install files for every os ready to install, command and file download. prompt9

Prompt 8: Detector + Policy Engine, Profile Ranking, Secure Storage + Next.js Landing Page

```
You are an expert full-stack systems engineer and UI/UX designer for privacy
tools. Continue building the "ShadowNet Agent" project precisely from all
previous prompts (1-7) and the detailed system diagram provided by the user.

Now implement two major parts:

**Part A: Core Control Plane Modules (Technical Backend)**

1. **Detector Subsystem** (real-time censorship detection)
   - Specify all detection methods with code skeletons (Go or TypeScript/Node
preferred for consistency).
   - Include: DNS poisoning check, SNI/TLS reset detection, HTTP injection
detection, QUIC/UDP drop detection, baseline connectivity tests.
   - Add jitter, rate limiting, and probe budget to avoid creating a stable
fingerprint.
   - Passive monitoring via sing-box logs/stats and OS network telemetry.

2. **Policy Engine (Deterministic)**
   - Rule-based decision maker that handles 80-90% of routine cases without
calling the LLM.
   - Define clear rules for switching profiles based on detection signals and
confidence scores.
   - Health scoring and ranking system for Profiles:
     - Each Profile has: ID, protocol, metadata, success_rate (EMA),
last_failure_reason, cooldown_until, health_score.
     - Ranking algorithm that prefers high-success, low-cooldown profiles while
enforcing diversity.

3. **Secure Local Store**
   - Design encrypted storage (using age encryption for Linux, Android Keystore
+ SQLCipher for Android).
   - Store: seed profiles, profile rankings/health data, bounded event ring
buffer (last 50 events only, no domains), trusted Signal contacts.
   - Wipe-on-suspicion mechanism for Inside version.
```

4. **Integration with shadownetd Supervisor**
   - How detector → policy engine → (optional LLM advisor) → supervisor applies atomic config reload to sing-box.

Part B: Next.js Landing Page & Documentation Site

Create a clean, modern, privacy-first Next.js 15+ (App Router) landing page and documentation site for ShadowNet Agent with these pages/sections:

- **Home / Hero Section**: Tagline "Local, Intelligent, Offline DPI Resistance for Iran" + subtle animated system diagram showing Inside/Outside versions.
- **Architecture Page**: Interactive or static diagram based on the user's system diagram (Data Plane + Control Plane), dual-version explanation, component flow.
- **Documentation / Docs**:
  - Installation guides for Linux, Android, Raspberry Pi.
  - Step-by-step setup wizard (CLI commands + screenshots).
  - How Inside and Outside versions work and how they coordinate via Signal.
  - Security model and threat model.
- **Download Page**:
  - Ready-to-install files for every OS:
    - Linux: Static binary (x86_64, arm64) + .deb/.rpm packages + systemd service file.
    - Android: APK (universal + arm64-v8a) for sideloading.
    - Raspberry Pi: Optimized ARM64 binary.
  - One-click download buttons with SHA256 checksums and verification instructions.
  - Direct commands (curl | sh) for Linux, Termux commands for Android.
- **Future / Roadmap**: Planned features (better mesh, synthetic data fine-tuning, etc.).
- **Footer**: Strong privacy statement, open-source license (AGPL-3.0), GitHub link (placeholder).

Technical requirements for the website:
- Fully static export possible (next export) for easy self-hosting.
- Dark mode by default, minimal JavaScript, no analytics/tracking.
- Responsive, fast, accessible.
- Use Tailwind CSS + shadcn/ui or similar for clean components.
- Include a simple "System Status Simulator" demo (fake detection + profile rotation animation) to show how the agent works.

Deliverables (well-structured):
1. Updated technical specification for Detector, Policy Engine, Profile Ranking, and Secure Store (with Go/TypeScript code skeletons).
2. Complete folder structure and key file list for the Next.js documentation site.
3. Page-by-page content outline with key sections and copy suggestions.
4. Download section specification: exact file names, commands, and verification steps for each platform.

```
5. Any necessary adjustments to the dual-version (Inside/Outside) design to
support the website.

Output everything in clear sections. Make the landing page professional,
trustworthy, and clearly targeted at users needing censorship circumvention in
Iran while remaining safe for developers.
```

Copy the entire prompt above and feed it to your local coding agent.

After processing Prompt 8, reply with "Prompt 9" — I will then deliver the final prompt covering:

- Full implementation skeletons for main loop + supervisor
- Testing framework with simulated DPI
- Build & release pipeline (including GitHub Actions for binaries and APK)
- Complete user guide content

This prompt nicely combines the missing technical core modules with the Next.js landing/documentation site you requested. We're getting very close to a complete, shippable project specification.

Write a promising ChiDio outline to produce a video and introduce the application and deliver it to the people inside Iran and expand the use and cooperation of developers around the world and anyone who believes in freedom of information in the age of artificial intelligence as a human right. The intelligent people of Iran are always friends and interested in modern technology and we aim to free the Iranian people from the tyranny of the dictator by calling on our friends and supporters to provide up-to-date information and free access to the Internet through an open source project. This project is free and available to everyone.

Gedanken für 13s

ShadowNet Agent – Video Production Outline
Title (English / Persian):
“ShadowNet: Your Local Key to a Free Internet – Open Source. Offline. Unstoppable.”

«عنوان فارسی: «شادونِت: کلید محلی اینترنت آزاد – اوپن‌سورس، آفلاین، تسخیرناپذیر

Video Length: 4:30–5:00 minutes (short, shareable, optimized for Telegram, WhatsApp, and X in Iran)

Style: Cinematic, dark-to-hopeful tone, minimal text on screen, fast cuts, subtle electronic music with Persian traditional undertones in the background. No faces of real users for security. Use AI-generated neutral animations + real screen recordings of the app.

Target Audiences (dual purpose):

1. People inside Iran (hope + immediate action)
2. Global developers, human-rights advocates, AI enthusiasts (call for contribution & support)

VIDEO OUTLINE (Scene-by-Scene)

0:00 – 0:25 | HOOK (Strong & Emotional)

- Black screen → sudden red “FILTERED” stamp that cracks and shatters.
- Quick montage: Iranian youth on phones trying to open Google, Wikipedia, YouTube → all blocked.

Voice-over (calm, confident Persian male/female bilingual narrator):
“در ایران ۱۴۰۵، اینترنت نه فقط فیلتر است... بلکه زندانی است.

- But what if your own device could outsmart the dictator’s firewall — locally, intelligently, without any server?”
- Logo reveal + tagline: ShadowNet Agent – The AI that never sleeps.

0:25 – 1:10 | THE PROBLEM (Real & Relatable)

- Real footage style (stock + animated): DPI in action (DNS poisoning, SNI reset, blackout screens).
- Statistics on screen (subtle, no source to avoid tracking):
    - “Blackouts in 2026 cut access to <5%”
    - “DPI now fingerprints even obfuscated protocols”
- Narrator: “The regime wants you in the dark. We built the light that lives inside your phone.”

1:10 – 2:20 | THE SOLUTION – ShadowNet Agent (Technical + Human)

- Clean 3D animation of the system diagram (exactly as designed in our prompts):
    - Inside version on phone/laptop
    - sing-box rotating protocols (Reality → Hysteria2 → TUIC)
    - Tiny Phi-4-mini LLM making decisions locally
    - Bluetooth mesh connecting neighbors in total blackout
    - Signal one-way config delivery from Outside helpers

Split screen:
Left: Iranian user's device (Inside version)

- Right: Global developer's laptop (Outside version generating fresh configs)

Narrator:

- "ShadowNet is two agents in one open-source project:
    - Inside: Your personal AI strategist that detects DPI in <5 seconds and rotates automatically.

Outside: Friends worldwide who generate fresh, unblockable configs and send them securely via Signal.

    - No central server. No logs. No cloud. Just your device + the global freedom network."

2:20 – 3:10 | WHY IT MATTERS – Human Rights in AI Age

- Powerful montage: Iranian students, doctors, journalists, programmers working at night with ShadowNet active.

Text overlays:
"Freedom of information is a human right in the age of AI."

- "Intelligent people of Iran have always been at the forefront of technology."

Narrator (emotional):
"This is not just a VPN. This is a declaration:

- No dictator can own the internet when the people own the code."

3:10 – 4:00 | CALL TO ACTION – Dual & Powerful

For people inside Iran:

- Screen shows real download buttons (Next.js landing page).
- "دانلود فوری – لینوکس | اندروید | Raspberry Pi"
- "دقیقه تا اینترنت آزاد ۱۰"
- "کامل با همسایه‌هایت وصل شو blackout حتی در"

For the world:

- "Developers & freedom lovers: This is AGPL-3.0 open source."
- "Join the Outside version. Generate configs. Help keep Iran online."
- GitHub link + Signal group for trusted helpers (anonymized).
- Final line: "One line of code from you = one hour of free internet for a family in Iran."

4:00 – 4:45 | CLOSING VISION

- Montage: Map of Iran lighting up node-by-node through Bluetooth mesh + global Signal connections.
- Text: "Free. Open. For everyone. Forever."

Final voice-over (strong, hopeful):
“ما ایرانی‌های هوشمند، با تکنولوژی مدرن، دیکتاتور را شکست می‌دهیم.

- ShadowNet is not just software. It’s the beginning of digital sovereignty for every human being.”
- Screen fades to: “Project is 100% free. No company. No government. Only people.”

4:45 – 5:00 | END CARD

- QR code to the Next.js landing page (shadon.et or GitHub)
- Hashtags (safe for Iran): #ShadowNet #اوپن‌سورس #اینترنت_آزاد
- Text: “Share this video. Build it. Use it. Freedom is code.”

Production Notes (for the team making the video):

- Music: Tense electronic → uplifting Persian fusion (royalty-free).
- Language: Primary Persian narration with English subtitles (for global reach).
- Length of Persian voice-over: ~4:40
- Visual style: Cyber-minimal, Matrix-inspired but hopeful (green code rain → blue freedom light).
- Call-to-action buttons must link to the exact download section of the Next.js site we designed in Prompt 8.

Distribution Strategy (built into the video):

1. First upload on secure channels (Telegram channels, Mastodon, X with VPN).
2. Encourage Iranian users to re-upload with Persian captions.
3. Global supporters: post on GitHub, Reddit (r/privacy, r/opensource), Hacker News, and AI freedom communities.
4. End screen asks: “If you believe in free information as a human right → fork, contribute, or become an Outside config provider.”

This outline is ready to hand to any video editor or AI video tool (Runway, Kling, etc.). It is emotionally strong, technically accurate, and perfectly balanced between inspiring Iranians and calling the world to action.

Would you like me to write the full Persian + English narration script next, or the shot-by-shot storyboard with exact timings and visual prompts for AI generation? Just say the word and we continue building.

Freedom is code. Let’s deliver it.

The landing page should contain links and buttons to download the installation file and recognize the operating system when opened and recommend the file. Also, explanations about the project's purpose and architecture, the project documentation page, and the installation guide are defined separately. On the landing page, links to project gates and the possibility of donuts for the project, as well as secure and anonymous cooperation of people with the project, and also that the license is mentioned. The responsibility for using the project lies with the people who install the project, and the goal is to bypass blockers of the free flow of information.

Prompt 10 for code agent

Prompt 10: Next.js Landing Page Implementation with Smart OS Detection + Full Project Polish

```
You are a senior full-stack developer and UI/UX expert specializing in
privacy-focused, high-trust open-source tools. Continue building the "ShadowNet
Agent" project from all previous prompts (1–9), including the dual
Inside/Outside architecture, system diagram, and video production outline.

Now implement the complete **Next.js 15+ Landing Page** (App Router) with the
following exact requirements:

### Core Features
- **Smart OS Detection & Download Recommendation**:
  - On page load, automatically detect the visitor's operating system (Linux,
Android, Windows, macOS, iOS, etc.) using navigator.userAgent and platform APIs.
  - Prominently recommend and highlight the most suitable download file for the
detected OS (e.g., Android APK for mobile users, Linux static binary for
desktop).
  - Show "Recommended for your system: [OS]" with a big primary download button.
  - Provide a full download section with tabs or cards for all supported
platforms:
    - Linux (x86_64 static binary, arm64, .deb, .rpm, systemd service)
    - Android (universal APK + arm64-v8a)
    - Raspberry Pi (ARM64 optimized)
    - Other (source code tarball)
  - Each download includes:
    - Direct download link (placeholder for now)
    - SHA256 checksum
    - One-line installation command (e.g., curl -L https://... | sh)
    - Verification instructions
  - Fallback: Manual OS selector if detection fails.

### Page Structure & Content (Clean, Dark Theme, Minimalist)
1. **Hero Section**:
   - Bold headline: "ShadowNet Agent — Local AI That Beats Censorship"
```

- Subheadline: "Offline. Intelligent. Unstoppable. Built for freedom of information."
- Big CTA buttons: "Download Now" (OS-aware) + "Learn How It Works"
- Subtle animated system diagram (Inside vs Outside versions)

2. **Project Purpose Section**:
   - Clear explanation: The goal is to bypass blockers of the free flow of information.
   - Emphasize: "Freedom of information is a human right in the age of artificial intelligence."
   - Mention dual versions: ShadowNet-Inside (for users in censored regions) and ShadowNet-Outside (for global supporters providing fresh configs).

3. **Architecture Overview**:
   - Embed or link to an interactive/simple diagram showing:
     - Data Plane (sing-box + TUN/VPN)
     - Control Plane (Detector, Policy Engine, Bounded LLM Advisor with Phi-4-mini, shadownetd)
     - Bluetooth Mesh + Signal coordination
   - Short technical summary (keep accessible, with "Read Full Architecture" link to docs).

4. **How It Works** (short cards):
   - Real-time DPI detection
   - Automatic protocol rotation (Reality, Hysteria2, TUIC, etc.)
   - Local LLM decisions (no cloud)
   - Offline resilience via Bluetooth mesh

5. **Downloads Section** (central and prominent):
   - OS detection + recommended file as described above
   - All platform cards with commands and checksums

6. **Documentation & Guides**:
   - Links to separate pages:
     - /docs (full project documentation)
     - /installation (step-by-step guide with screenshots)
     - /architecture (detailed technical deep-dive)
   - "Installation in under 10 minutes" teaser

7. **Get Involved & Secure Cooperation**:
   - Ways to contribute anonymously and securely:
     - Run the Outside version and send configs via Signal
     - Report DPI fingerprint changes (anonymously)
     - Code contributions via GitHub (AGPL-3.0)
     - Donate (anonymous cryptocurrency options – add placeholder wallet addresses)
   - Strong privacy note: "All cooperation can be done anonymously. No personal data required."

```
8. **License & Responsibility**:
   - Clearly state: "Licensed under AGPL-3.0 — fully open source and free for
everyone."
   - Disclaimer: "The responsibility for using this project lies entirely with
the individuals who install and run it. ShadowNet is a tool to bypass blockers
of the free flow of information. Use at your own risk and in accordance with
your local laws."

9. **Footer**:
   - Links: GitHub, Docs, Installation, Architecture, Video (from previous
outline)
   - "Built by people who believe freedom of information is a human right."
   - No tracking, no analytics, dark mode default.

### Technical Requirements
- Next.js 15+ App Router + TypeScript
- Tailwind CSS + shadcn/ui components for clean, accessible UI
- Fully static export capable (`next export` or output: 'export')
- Dark theme by default, modern cyber-minimalist design
- Mobile-first and fully responsive
- Zero external dependencies that could track users (no Google fonts if
possible, self-host everything)
- Fast loading (<2s on slow connections)
- SEO-friendly meta tags with focus on "censorship circumvention", "offline DPI
bypass", "open source Iran internet freedom"

Deliverables:
1. Complete folder structure for the Next.js project.
2. Key page code skeletons (page.tsx for /, /docs, /installation,
/architecture).
3. Smart OS detection hook/component with recommendation logic (ready-to-use
code).
4. Download section component with platform cards and commands.
5. Exact disclaimer and license text blocks.
6. Suggestions for hosting (GitHub Pages, IPFS, self-hosted) for maximum
accessibility inside Iran.

Make the landing page trustworthy, professional, emotionally resonant, and
clearly targeted at both Iranian users seeking freedom and global supporters
who want to help. Prioritize clarity, security, and ease of use.
```

Copy the entire prompt above and feed it to your local coding agent.

After processing Prompt 10, reply with "Prompt 11" if you want to continue the chain (e.g., full implementation of the main agent loop, testing framework, or video narration script).

This prompt fully incorporates all your latest requirements for the landing page: OS detection with smart recommendations, separate docs/installation pages, donation/cooperation options, license mention, and clear responsibility disclaimer.

We're building something meaningful. Let me know when you're ready for the next step.

A comprehensive and complete test of the performance of the entire project prompt 11

Prompt 11: Comprehensive Testing Framework for ShadowNet Agent – Full Project Performance, Security & Real-World Validation

```
You are a senior QA engineer and security researcher specializing in censorship
circumvention tools and offline-first systems. Continue building the "ShadowNet
Agent" project from all previous prompts (1–10), including the dual
Inside/Outside architecture, sing-box core, bounded LLM advisor (Phi-4-mini
Q4_K_M), detection/policy engine, Bluetooth mesh, Signal config distribution,
secure storage, and the Next.js landing page.

Now design and implement a **comprehensive, complete, and production-ready
testing framework** that covers every aspect of the entire project.

### Testing Goals
- Validate that the agent reliably detects Iranian-style DPI and maintains
connectivity under heavy censorship simulation.
- Ensure the system stays within strict resource limits (RAM <800MB, fast
switching <5s).
- Verify security, anonymity, and seizure-resistance properties.
- Test both Inside and Outside versions and their coordination.
- Confirm the Next.js landing page works correctly (OS detection, downloads,
etc.).
- Provide reproducible tests that a solo developer can run locally.

### Required Test Categories (Must Cover All)

1. **Unit Tests**
   - Detector subsystem (DNS poisoning, SNI reset, HTTP injection, QUIC drop,
baseline checks).
   - Policy Engine rules and profile ranking/health scoring logic.
   - LLM Advisor: JSON schema validation, fallback behavior when output is
invalid.
   - Profile generation and mutation logic (bounded parameter changes).
   - Secure storage encryption/decryption and wipe-on-suspicion.

2. **Integration Tests**
   - Full detection → policy → LLM advisor → config generation → sing-box
hot-reload cycle.
```

- End-to-end Inside agent loop under simulated network conditions.
- Outside version config generation and Signal bundle creation/validation.
- Bluetooth mesh simulation (peer discovery, encrypted proxy exchange, hop-by-hop forwarding).

3. **Performance & Resource Tests**
   - RAM/CPU usage benchmarks (target: 400-600MB peak for Inside version).
   - Decision latency: detection + LLM inference + config switch <5 seconds (95th percentile).
   - Battery impact on Android (foreground service throttling).
   - Rotation stability: 1000+ automatic switches without crashes or leaks.
   - Hot-reload downtime measurement (<2-5 seconds).

4. **Censorship Simulation Tests (Most Critical)**
   - Build a realistic DPI simulator (using tools like iptables, tc, dnsmasq, or custom Go/Python proxy).
   - Simulate real 2026 Iranian DPI behaviors:
     - DNS poisoning to 10.10.34.x blackhole
     - SNI-based RST after ClientHello
     - HTTP 403 injection
     - Selective QUIC/UDP drop
     - Whitelist-only access (some domains allowed, others blocked)
     - Full blackout mode (zero external connectivity)
   - Test agent recovery time and success rate across 50+ simulation scenarios.
   - Measure effectiveness of each protocol (Reality, Hysteria2, TUIC, ShadowTLS, DNS fallback).
   - Test diversity: does frequent mutation of SNI/uTLS/shortId prevent fingerprint blocking?

5. **Security & Privacy Tests**
   - No leakage of visited domains, IP addresses, or user data in logs/storage.
   - Seizure resistance: simulate device inspection – can sensitive data be recovered?
   - Mesh communication: verify end-to-end encryption and anonymity (no identity leakage).
   - Signal bundle: validate encryption, signing, and safe import on Inside version.
   - LLM isolation: ensure LLM never receives secrets.

6. **Cross-Platform & Deployment Tests**
   - Linux (x86_64, arm64), Android APK, Raspberry Pi.
   - Next.js landing page: OS detection accuracy, download links, static export.
   - Installation flows: CLI commands, systemd service, Android foreground service.

7. **End-to-End Real-World Simulation**
   - Run Inside + Outside agents together.
   - Outside generates fresh configs → sends via simulated Signal → Inside receives and applies.

```
   - Full blackout test with Bluetooth mesh between multiple simulated devices.
   - Long-running stability test (24-48 hours under varying censorship levels).

### Deliverables (Practical & Ready-to-Use)
1. **Test Framework Architecture**:
   - Recommended tools: Go testing (for core), Jest/Playwright (for Next.js),
custom DPI simulator in Go/Python.
   - Directory structure: `/tests/unit`, `/tests/integration`,
`/tests/simulation`, `/tests/performance`.

2. **Complete Test Suites** (code skeletons for key tests):
   - DPI Simulator implementation (with configurable scenarios).
   - Automated benchmark scripts for latency, resource usage, and switch success
rate.
   - Security audit checklist with test cases.

3. **Testing Pipeline**:
   - CI/CD suggestions (GitHub Actions) for automated unit/integration tests.
   - Manual testing guide for real Iranian conditions (when safe).

4. **Success Metrics** (define clear pass/fail criteria):
   - 95%+ success rate in maintaining connectivity under simulated DPI.
   - Zero sensitive data leaks.
   - Resource limits respected.
   - Seamless Inside ↔ Outside coordination.

5. **Edge Cases & Failure Modes**:
   - LLM crash/fallback behavior.
   - Zero seed configs left.
   - Low battery / low RAM scenarios.
   - Malicious config injection attempt.

Make the testing framework comprehensive yet practical for a solo developer.
Include setup instructions to run all tests locally on a Linux machine.
Emphasize realism to 2026 Iranian DPI conditions while keeping everything safe
and reproducible.

Output in clear, structured sections with code examples, test case tables, and
ready-to-implement skeletons. This should be the final validation layer that
gives confidence the entire ShadowNet Agent project is robust, effective, and
ready for real use.
```

Copy the entire prompt above and feed it to your local coding agent.

After processing Prompt 11, reply with "Prompt 12" if you want to continue (e.g., full main loop implementation, video narration script, or release checklist).

This prompt delivers exactly what you asked for: a comprehensive and complete testing strategy for the entire project — performance, security, censorship resistance, cross-platform behavior, and real-world simulation. It ties together all previous components into a verifiable, production-grade system.

We're very close to having a complete, buildable, and testable open-source project. Let me know when you're ready for the next step.

The Android and Apple mobile app must work in the background in such a way that no one can easily know it is installed on the device, or it must be very difficult and not possible on site.

Prompt 12

Prompt 12: Stealth Background Operation for Mobile Apps – Android & iOS (Apple) – Maximum Concealment

```
You are an expert in mobile security, anti-forensics, and high-risk censorship
circumvention tools. Continue building the "ShadowNet Agent" project from all
previous prompts (1–11), including the dual Inside/Outside architecture,
sing-box core, bounded LLM advisor, detection/policy engine, Bluetooth mesh,
Signal config sharing, Next.js landing page, and comprehensive testing
framework.

Now add critical stealth and anti-forensics requirements specifically for the
**mobile versions**:

### Hard Requirements for Mobile Stealth
The Android and iOS versions must be designed so that:
- It is extremely difficult for anyone performing a casual or on-site device
inspection (family member, security forces, or basic forensic check) to discover
that ShadowNet Agent is installed and running.
- The app must continue operating fully in the background (detection, profile
rotation, sing-box tunneling, Bluetooth mesh scanning, Signal checking) even
when the app is not in foreground.
- No obvious icons, notifications, or visible traces should appear by default.

### Android-Specific Stealth Implementation (Primary Focus – Higher Risk
Environment)
1. **Installation & Concealment Techniques**:
   - Support "hidden installation" methods: sideload as a seemingly innocent app
(e.g., disguised as a calculator, flashlight, weather app, or system service).
   - Ability to hide the app icon completely after installation (using launcher
hiding techniques or overlay methods).
```

- Use of Android Work Profile or secondary user profiles where possible for isolation.
- Option to rename package name at build time (e.g., com.system.service.update or similar innocuous names).
- Disable all default notifications; use silent foreground service with minimal or no persistent notification (exploit Android exemptions for VPN services).

2. **Background Execution**:
   - Run as a persistent foreground VPN service (using VpnService API) so the system keeps it alive even after app is swiped away.
   - Implement aggressive battery optimization bypass (using ignoreBatteryOptimizations + power whitelist tricks where allowed).
   - Use WorkManager + AlarmManager for periodic detection/LLM decisions when the service is throttled.
   - Maintain sing-box in a sidecar or embedded native process.

3. **Anti-Forensics & On-Device Hiding**:
   - No app name or icon visible in launcher by default (user can choose "stealth mode" during setup).
   - Hide from recent apps list and app switcher.
   - Minimal or encrypted app data folder names.
   - Self-defense: if suspicious access detected (e.g., unusual ADB/debug mode), trigger graceful shutdown or data wipe.
   - Option for "decoy mode": app appears as a harmless utility with fake UI while real agent runs in background.

iOS (Apple) Stealth Implementation (More Restricted)
1. **Background Operation**:
   - Use Network Extension framework (Packet Tunnel Provider) for VPN-like tunneling with sing-box.
   - Leverage Background Modes (background fetch, VoIP, location if needed) to keep the extension alive.
   - Limited persistence due to iOS restrictions — design fallback strategies (e.g., user must occasionally open app to "refresh").

2. **Concealment**:
   - Disguise app as a legitimate-looking utility (calculator, notes, health tracker, etc.).
   - No persistent notifications.
   - Use app groups and shared containers carefully to minimize footprint.
   - On-site inspection resistance: app should not appear suspicious in Settings > VPN or General > Profiles & Device Management.

General Mobile Requirements Across Both Platforms
- **Stealth Mode Toggle**: During initial setup, user can enable "Maximum Stealth" which:
  - Hides icon
  - Renames package/bundle ID (Android)

```
  - Removes all visible traces
  - Runs completely silent (no sounds, no banners)
- **Access Method**: Provide a secret gesture, code, or hidden launcher (e.g., specific pattern in calculator UI) to access real configuration UI only when needed.
- **Resource Awareness**: Extremely conservative CPU/RAM usage in background to avoid suspicion from battery drain or heat.
- **Wipe-on-Danger**: Quick emergency wipe of configs and logs if user triggers panic mode (e.g., wrong password 5 times or specific gesture).
- **Compatibility**: Must work on non-rooted Android and stock iOS (no jailbreak required).

### Integration with Existing Architecture
- The core shadownetd logic, detector, policy engine, LLM advisor, and sing-box must run unchanged in the background mobile process.
- Bluetooth mesh and Signal checking must continue silently.
- Update the testing framework (from Prompt 11) to include mobile-specific stealth tests:
  - Visibility audit (does the app appear in launcher, settings, notifications?).
  - Battery and performance forensics simulation.
  - On-site inspection scenarios (what an examiner would see in 5–10 minutes).

### Deliverables
1. Detailed stealth architecture for Android and iOS, including recommended disguise strategies and technical tricks.
2. Updated mobile app architecture diagram showing how background service interacts with sing-box and LLM.
3. Step-by-step implementation guide for "Maximum Stealth Mode" on both platforms.
4. Code skeletons for:
   - Android: Stealth foreground VPN service + icon hiding + decoy UI.
   - iOS: Network Extension with background persistence + disguise.
5. Updated security and anti-forensics section in the project documentation.
6. Additions to the comprehensive testing framework: new test cases for stealth effectiveness and background reliability.
7. Warnings and trade-offs: explain limitations (especially on iOS) and legal/ethical notes regarding responsibility (as defined in previous prompts).

Output in clear, structured sections with practical code examples and realistic assessments for 2026 high-censorship environments. Prioritize maximum concealment while maintaining functionality and stability. Keep everything practical for local development and open-source distribution.
```

Copy the entire prompt above and feed it to your local coding agent.

After you get the output and integrate it, reply with "Prompt 13" if you want to continue the chain.

This prompt directly addresses your new requirement for strong stealth/background operation on Android and Apple devices, making the app very difficult to detect during on-site inspections while keeping full functionality. It integrates cleanly with the existing dual-version design and testing framework.

We're building a serious, responsible tool for high-risk situations. Let me know when you're ready for the next prompt.

After processing Prompt 8, reply with "Prompt 9" — I will then deliver the final prompt covering:

Full implementation skeletons for main loop + supervisor
Testing framework with simulated DPI
Build & release pipeline (including GitHub Actions for binaries and APK)
Complete user guide content

prompt

Prompt 12: Stealth Background Operation for Mobile Apps – Android & iOS (Apple) – Maximum Concealment (Updated with 2026 Realities)

```
You are an expert in mobile anti-forensics, high-risk circumvention tools, and
Android/iOS background execution under strict platform restrictions (2026
Android 15+/16 and iOS 26+). Continue building the "ShadowNet Agent" project
from all previous prompts (1–11), including dual Inside/Outside architecture,
sing-box core, bounded Phi-4-mini LLM advisor, detector/policy engine, Bluetooth
mesh, Signal config sharing, Next.js landing page, and comprehensive testing
framework.

Incorporate current 2026 platform realities:
- Android: VPN key icon in status bar is harder to hide on stock ROMs;
foreground service notification is often mandatory for VpnService; launcher
hiding is possible via Settings or custom tricks; battery optimizations are
stricter.
- iOS: Network Extension (Packet Tunnel Provider) has improved background
persistence but still limited; no easy icon hiding without disguise; Background
Security Improvements and privacy indicators are active.

### Hard Requirements for Mobile Stealth (Inside Version)
The Android and iOS versions must make it **extremely difficult** for
casual/on-site inspection (security forces, family, basic forensic check without
advanced tools) to discover that ShadowNet is installed and actively running.
The agent must continue full operation (real-time DPI detection, automatic
profile rotation, sing-box tunneling, silent Bluetooth mesh scanning, Signal
config polling) in the background with minimal visible footprint.
```

Android Stealth Implementation (Primary – Higher Risk)

1. **Disguise & Icon Hiding**:
   - Build-time package name randomization or innocuous names (e.g., com.android.system.updater, com.google.services.sync).
   - Support "decoy mode": App launches as a harmless utility (calculator, flashlight, notes, or weather app) with fake UI. Real controls accessible only via secret gesture/code/pattern.
   - Hide app icon from launcher and recent apps list (use Android's hide-apps API or overlay tricks where possible; fallback to user manual hiding via Settings > Home Screen).
   - Option for complete icon removal after first setup.

2. **Background Execution & Persistence**:
   - Run as a **foreground VPN Service** using Android VpnService API (required for system-wide TUN routing with sing-box). This keeps the process alive even if swiped away.
   - Minimize or suppress the persistent notification:
     - Use low-priority channel + empty/small icon.
     - Exploit exemptions for VPN services.
     - Provide option to hide VPN key icon (note limitations on stock Android/Pixel/Samsung; recommend Samsung Good Lock-style tricks or custom ROM awareness).
   - Aggressive Doze/battery optimization bypass: request ignoreBatteryOptimizations, use WorkManager + exact alarms for periodic wake-ups when throttled.
   - Embed sing-box as native library or sidecar process for stable tunneling.

3. **Anti-Forensics Features**:
   - No visible traces in app list, settings, or notifications by default in "Maximum Stealth Mode".
   - Encrypted/obfuscated data directories.
   - Panic wipe: specific gesture or wrong access code 5+ times triggers graceful shutdown + deletion of configs/seeds/logs.
   - Self-defense: Detect ADB/debug mode or unusual app inspector activity and enter decoy-only mode.

iOS Stealth Implementation (More Restricted)

1. **Background Operation**:
   - Use **Network Extension Packet Tunnel Provider** for sing-box integration (system-level VPN).
   - Enable Background Modes (background fetch, VoIP if acceptable, or location as fallback – with user consent warning).
   - Due to iOS 26 restrictions, design for occasional user "refresh" open (e.g., open decoy app once per day) while extension runs in background.

2. **Concealment**:
   - Disguise the app as a legitimate utility (calculator, health tracker, notes app) with fake UI.
   - No persistent notifications or banners.

  - Minimize visibility in Settings > VPN & Device Management.
  - Secret access to real UI via hidden gesture or code in the decoy interface.

Cross-Platform Requirements
- **Maximum Stealth Mode** during initial setup (via wizard on Next.js landing or in-app):
  - Enables all hiding features.
  - Runs completely silent (no sounds, minimal CPU to avoid heat/battery suspicion).
- **Access Control**: Secret gesture/pattern to reveal real configuration UI only when needed.
- **Resource Conservatism**: Strict limits on background CPU/RAM to prevent suspicion from battery drain or device heating. Integrate with mobile testing framework.
- **Compatibility**: Non-rooted Android and stock (non-jailbroken) iOS.

Integration with Core Architecture
- shadownetd supervisor, detector, policy engine, LLM advisor, and sing-box run unchanged inside the background service/extension.
- Bluetooth mesh and Signal checks operate silently.
- Update testing framework (Prompt 11) with new mobile stealth test cases:
  - Visibility audit (launcher, settings, notifications, status bar).
  - Forensic simulation (what an on-site examiner sees in 5–10 minutes).
  - Battery/performance forensics.
  - Panic wipe effectiveness.

Deliverables (Practical for Solo Development)
1. Updated mobile architecture diagram showing background service/extension interaction with sing-box/LLM.
2. Detailed implementation guide for Android (foreground VPN service + disguise + icon hiding) and iOS (Network Extension + decoy UI).
3. Code skeletons:
   - Android: Stealth VpnService class, decoy Activity, notification minimization.
   - iOS: PacketTunnelProvider with background handling + disguise logic.
4. "Maximum Stealth Mode" setup flow (integrated with Next.js installation guide).
5. Updated documentation sections: security/anti-forensics, mobile deployment, limitations & trade-offs (especially iOS persistence limits).
6. Additions to comprehensive testing framework: stealth-specific test cases and success criteria.
7. Clear warnings: Trade-offs (battery, visibility on certain devices), legal responsibility disclaimer (user bears all risk), and ethical note on use for bypassing information blockers.

Output in well-structured sections with realistic 2026 platform assessments, code examples, and practical recommendations. Prioritize maximum on-site

```
concealment while preserving full functionality and stability. Ensure the
design remains open-source friendly under AGPL-3.0.
```

Copy the entire prompt above and feed it to your local coding agent.

This updated Prompt 12 incorporates real 2026 platform behaviors (VPN icon/notification challenges on Android, iOS background restrictions) while staying faithful to the stealth requirements you described.

After processing this prompt, reply with "Prompt 13" if you want to continue the chain (e.g., full main loop + supervisor skeletons, build pipeline, or complete user guide content as originally planned after Prompt 8).

We're now covering the critical mobile stealth layer needed for real-world use inside high-risk environments. Let me know when you're ready for the next step.

20 sources

prompt to create a logo für icon

2 / 2

Prompt 13: Logo & App Icon Design for ShadowNet Agent

```
You are a professional logo designer and brand identity expert specializing in
privacy, security, and censorship circumvention tools. Create a complete,
modern, and meaningful visual identity for "ShadowNet Agent" — an open-source,
offline, AI-powered tool that helps users in censored regions (especially Iran)
bypass DPI and maintain free internet access.

### Core Requirements for the Logo & Icon
- **Concept**: The logo must symbolize:
```

  - Stealth / invisibility / shadow
  - Intelligence (AI / local LLM)
  - Connectivity / breaking chains / freedom of information
  - Resilience against powerful blocking systems
  - Minimalist, professional, and trustworthy (not childish or overly aggressive)
  - Work perfectly in both light and dark themes (the project uses dark mode by default)

- **Main Elements to Consider**:
  - A stylized "S" or "SN" monogram
  - Subtle network / node / mesh pattern (representing Bluetooth mesh and global Outside helpers)
  - Shadow effect or negative space that implies "hidden in plain sight"
  - A small shield or key element representing protection and unlocking access
  - Optional subtle circuit / neural network lines to hint at the local AI (Phi-4-mini) without being too techy
  - Avoid: skulls, guns, obvious "hacker" imagery, national flags, or anything that could be seen as political provocation

- **Style**:
  - Modern minimalist vector style (clean lines, geometric)
  - Cyber-minimalist with a touch of hope (dark blues, deep purples, neon cyan accents optional)
  - Highly scalable: must look sharp at 512x512 (app icon), 256x256, 64x64, and as favicon

Deliverables
1. **Primary Logo**:
   - Horizontal version (with "ShadowNet" text)
   - Stacked/vertical version
   - Icon-only version (for app launcher)

2. **App Icon Specifications**:
   - Square icon (1024x1024 px base) optimized for Android Adaptive Icon and iOS
   - Stealth-friendly: should not scream "VPN" or "hacking tool" — look like a neutral utility or system app
   - Variants:
     - Standard dark version
     - Light version
     - Monochrome version (for notifications or minimal UI)

3. **Color Palette**:
   - Primary: Deep navy (#0A1428) or charcoal
   - Accent: Electric cyan (#00F0FF) or soft purple (#8B5CF6) for "connection breaking through"
   - Neutral grays for background

4. **Usage Guidelines**:

```
   - Where to use the full logo vs icon-only
   - Minimum size and clear space rules
   - How it integrates with the Next.js landing page (dark theme)

5. **Text Suggestions** (if used):
   - "ShadowNet" in a modern sans-serif font (e.g., Inter, Satoshi, or similar)
   - Optional small tagline under logo: "Local • Intelligent • Free"

Provide:
- Detailed textual description of the logo concept (so it can be fed to an image
generation tool like Grok Imagine, Midjourney, or Flux).
- 3 different concept variations with explanations.
- Ready-to-use prompt for AI image generation (highly detailed, vector-style,
suitable for professional logo).
- Suggestions for favicon, social media assets, and documentation site usage.

Make the logo elegant, memorable, and professional enough to build long-term
trust while remaining discreet for users in high-risk environments.
```

Copy the entire prompt above and feed it to your local coding agent (or directly to an image generation tool if your agent supports it).

This prompt is optimized to produce a strong, meaningful, and stealth-appropriate logo/icon for ShadowNet Agent.

After you generate the logo concepts, you can reply with "Prompt 14" if you want to continue the chain (for example, full main agent loop implementation or complete release checklist).

Would you like me to adjust this prompt (e.g., make it more minimal, add Persian elements, or focus only on the app icon)? Just say so and I'll refine it immediately.

Create a separate Donut Page where users can create a Revolte account according to the conditions in the photo and earn up to $200 by using their good and free services. We will put our link on the page. Also, a QR code for donating to the project. Prompt 13

Prompt 13: Dedicated Donations / Support Page – "Support ShadowNet" with Revolut Referral Integration + QR Code

```
You are a senior Next.js developer and UX designer for privacy-focused
open-source projects. Continue building the ShadowNet Agent website from all
previous prompts (especially Prompt 10 for the main landing page).
```

Now create a **separate, dedicated page** at route `/support` or `/donate` (with clear navigation link from the main landing page and footer).

Page Purpose
- Provide ways for users (especially global supporters) to financially support the project anonymously and securely.
- Integrate a Revolut referral program to let users create a Revolut account, complete simple requirements, and earn up to $200 (or equivalent in their currency) while indirectly supporting the project via your referral link.
- Include a prominent QR code for direct donations to the project (cryptocurrency or other anonymous methods).
- Keep the tone professional, transparent, and aligned with the project's mission: freedom of information as a human right, fully open-source, no central control.

Page Structure & Content (Dark theme, consistent with main landing page)

1. **Hero Section**
   - Headline: "Support ShadowNet – Keep the Light On for Free Internet"
   - Subheadline: "Every contribution helps maintain and improve the tool that bypasses censorship. You can support us while earning rewards yourself."
   - Two big cards side-by-side:
     - Card 1: "Earn up to $200 with Revolut"
     - Card 2: "Direct Anonymous Donation"

2. **Revolut Referral Section** (Main Feature)
   - Clear explanation: "Create a free Revolut account using our referral link. Complete simple steps (add money + make qualifying transactions) and earn up to $200 welcome bonus (amount varies by region and current promotion – check Revolut for latest terms)."
   - Prominent button: **"Create Revolut Account & Earn up to $200"** that links to your Revolut referral link (use a placeholder like `https://revolut.com/referral/?code=YOURCODE` – make it easily editable).
   - Step-by-step instructions (based on typical Revolut referral flow):
     1. Click the link and sign up.
     2. Verify your identity.
     3. Add money to your account.
     4. Make the required qualifying transactions (usually 1–3 small purchases).
     5. Receive your bonus.
   - Disclaimer: "Terms and bonus amounts are set by Revolut and may vary. We only provide the referral link as one way to support the project indirectly."
   - Note the benefit to the project: "Using our link helps sustain development of ShadowNet."

3. **Direct Donation Section**
   - Headline: "Direct Support via Cryptocurrency (Fully Anonymous)"
   - Display a large, scannable **QR code** for the project's donation wallet (use a placeholder Bitcoin, Monero, or USDT address – make it configurable).
   - Text below QR: "Scan to donate BTC / XMR / USDT or copy the address."

```
   - List of supported cryptocurrencies with addresses (in a clean table or
cards).
   - Optional: "One-time or recurring – every satoshi helps keep ShadowNet
updated against new DPI techniques."

4. **Other Support Options**
   - Run the Outside version and share fresh configs via Signal.
   - Contribute code on GitHub (AGPL-3.0).
   - Spread the word (share the landing page or video).

5. **Transparency & Responsibility**
   - Clear statement: "ShadowNet is 100% free and open-source. All donations go
directly to development, server costs for testing, and maintaining config seeds.
No company or organization takes a cut."
   - Reminder of the main project disclaimer: "Responsibility for using
ShadowNet lies with the user. The goal is bypassing blockers of the free flow of
information."

### Technical Requirements
- Next.js 15 App Router page: `/support` or `/donate`
- Fully responsive and accessible.
- Use the same Tailwind + shadcn/ui design system as the main landing page.
- Make the Revolut referral link and donation addresses easy to update via a
config file or environment variable.
- QR code generation: Use a React component (e.g., qrcode.react) that generates
the QR from the donation address on the server or client side.
- SEO-friendly title: "Support ShadowNet Agent – Donate or Earn with Revolut"
- Add breadcrumb navigation: Home → Support

### Deliverables
1. Complete page code skeleton (`app/support/page.tsx` or similar) with all
sections.
2. Reusable components: RevolutReferralCard, DonationQRCard.
3. Updated navigation (add "Support" link to main layout and footer).
4. Exact disclaimer texts for Revolut referral and donations.
5. Suggestions for the actual Revolut referral link and donation wallet setup
(recommend Monero or BTC for maximum anonymity).

Make the page trustworthy, non-intrusive, and clearly separate from the main
download-focused landing page. Emphasize that supporting the project is optional
and users can still use everything for free.

Output in clean, production-ready structure with code examples.
```

Copy the entire prompt above and feed it to your local coding agent.

This prompt creates exactly what you asked for: a separate support/donate page with Revolut account creation + earning opportunity (using your referral link) and a QR code for project donations.

After processing, you can reply with "Prompt 14" to continue the chain.

Note: Revolut referral bonuses vary by country and time — the page should clearly state that users must check current terms on Revolut's site.

Let me know if you want any adjustments (e.g., change route to `/donate`, add more payment options, or make the Revolut section more prominent).

20 sources

آیا میتوانیم سیستم خودمان را از بیرون در شرایط واقعی تست کنیم راهی سراغ دارید . تست واقعی و متریک عملکرد ها معناست.

Gedanken für 18s

بله، کاملاً ممکنه سیستم ShadowNet رو در شرایط واقعی DPI بگیریم (از بیرون) تست کنیم و متریک‌های عملکرد واقعی ایران.

اما نکته کلیدی: تست واقعی DPI فقط از داخل شبکه ایران ممکنه، چون DPI در gatewayهای TIC/IPM از ایران اعمال می‌شه. رو حس کنی، ولی می‌تونی از بیرون متریک‌ها رو جمع‌آوری و تحلیل کنی با کمک کاربران داخل DPI بیرون نمی‌تونی مستقیم (داوطلب و امن).

(راه‌های عملی و امن برای تست واقعی (به ترتیب اولویت و عملی بودن در ۱۴۰۵/۲۰۲۶

1. (بهترین و واقعی‌ترین روش) Signal برنامه بتا تست با داوطلبان داخل + گزارش متریک از طریق
    - کاربران داخل ShadowNet-Inside (در حالت Stealth) رو نصب می‌کنن.

- Prompt 11 که در testing framework ایجنت خودش متریک‌های کلیدی رو جمع می‌کنه (از همون طراحی کردیم):
    - نرخ موفقیت اتصال (success rate) هر پروتکل (Reality, Hysteria2, TUIC, ShadowTLS, DNS Tunnel)
    - زمان تشخیص بلاک (detection latency < 5s)
    - زمان سوئیچ کانفیگ (hot-reload time)
    - uptime کامل در blackout
    - تعداد rotation موفق در هر ساعت
    - مصرف رم/باتری
    - موفقیت Bluetooth Mesh در blackout
- بدون سایت بازدیدشده، بدون شناسه کاربر) ذخیره IP، بدون) onymized JSONمتریک‌ها به صورت ان می‌شن.
- امن بفرسته (یا Outside به یک حساب Signal کاربر داخل می‌تونه با یک دکمه ساده، متریک‌ها رو از طریق QR کد بزنه).
- خاص ISP در Reality متریک‌ها رو جمع‌آوری، تحلیل و داشبورد می‌سازه (مثلاً موفقیت Outside version چقدر بوده؟).

2. چطور شروع کنیم؟
    - داوطلب جذب کن (circumvention از طریق کانال‌های امن موجود (تلگرام/سیگنال گروه‌های.
    - استفاده می‌کردن Psiphon/Snowflake اول ۱۰–۲۰ نفر تست‌کننده اولیه (از کسانی که قبلاً).
    - بعد از تست اولیه، متریک‌ها رو عمومی کن (مثل گزارش‌های OONI).
3. ادغام با OONI Probe (طلایی‌ترین روش اندازه‌گیری عمومی)
    - و DNS poisoning، DPI الان هم در ایران بیش از ۴۰ میلیون اندازه‌گیری داره و دقیق OONI Probe SNI reset رو اندازه می‌گیره.
    - خروجی بده (این OONI رو طوری طراحی کنی که متریک‌های خودش رو در فرمت ShadowNet می‌تونی می‌تونیم اضافه کنیم Prompt 11 کار رو در).
    - کاربران داخل OONI Probe رو هم اجرا کنن ← داده‌ها عمومی و قابل مقایسه با ابزارهای دیگه (Psiphon, Tor Snowflake, Lantern) می‌شن.
    - رو مانیتور کنی و ببینیم OONI Explorer (explorer.ooni.org/country/IR) از بیرون می‌تونی چقدر بهتر از بقیه عمل کرده ShadowNet.
4. مدل Psiphon Conduit (خارجی‌ها کمک مستقیم می‌کنن)
    - رو اجرا می‌کنن و کانفیگ تازه تولید Outside ShadowNet یا نسخه Conduit هاOutside helper می‌کنن.
    - داخل تست می‌کنه و موفقیت/شکست رو گزارش می‌ده.
    - خیلی استفاده می‌شه و کاملاً عملی و امن هست Snowflake و Psiphon این روش الان در ۲۰۲۶ برای.
5. (تست شبیه‌سازی پیشرفته + اعتبار سنجی واقعی (ترکیبی
    - اول با DPI Simulator (که در Prompt 11 تست محلی می‌کنی طراحی کردیم).
    - می‌کنی تا دقیق‌تر بشه calibrate رو Simulator بعد نتایج واقعی از داوطلبان داخل می‌گیری و.
    - اینطوری ۸۰٪ تست‌ها رو از بیرون انجام می‌دی و فقط اعتبار نهایی رو از داخل می‌گیری.

متریک‌های عملکردی که می‌تونیم واقعی اندازه بگیریم

- Success Rate per Protocol: مثلاً Reality در ۸۵٪ موارد ۳۰ دقیقه زنده می‌مونه یا نه.

- Average Detection Time: چقدر طول می‌کشه بلاک تشخیص داده بشه.
- Switch Success Rate: بعد از تشخیص، چند درصد مواقع سوئیچ موفق بوده.
- Uptime in Blackout: درصد زمانی که حتی در blackout کامل اینترنت داشته.
- Mesh Effectiveness: چقدر ترافیک رو بین همسایه‌ها منتقل کرده Bluetooth Mesh، در blackout.
- Resource Usage: در شرایط واقعی CPU، رم، باتری.
- Fingerprint Survival: بعد از آپدیت DPI رژیم، کانفیگ جدید چقدر سریع زنده می‌شه.

نکات امنیتی خیلی مهم

- هیچ‌وقت متریک خام (با IP جمع نکن (یا لاگ ترافیک.
- همه گزارش‌ها opt-in و رمزنگاری end-to-end باشه.
- کاربران داخل هیچ ریسکی نکنن (فقط داوطلب آگاه).
- همون (مسئولیت کامل با کاربره disclaimer که قبلاً گذاشتیم).

نتیجه‌گیری:

بله، کاملاً می‌تونیم تست واقعی انجام بدیم و متریک‌های معنادار بگیریم — بهترین راه بتا تست با داوطلبان داخل + گزارش سیگنال + الان استفاده می‌کنن و در گزارش‌های ۲۰۲۶ OONI و Psiphon, Tor هست. این روش دقیقاً همون چیزیه که Outside داشبورد هم بهش اشاره شده.

اگر بخوای، می‌تونم Prompt 14 رو بنویسم برای:

- پیاده‌سازی سیستم جمع‌آوری متریک anonymized
- داشبورد ساده برای Outside version
- راهنمای جذب داوطلب امن

یا بگو دقیقاً کدوم بخش رو اول می‌خوای "Prompt 14" بگو.

ما می‌تونیم این سیستم رو واقعاً تست کنیم و بهتر از ابزارهای فعلی کنیم. آماده‌ای؟

45 sources

Prompt 14

Prompt 14: Real-World Testing & Anonymized Metrics Collection System for ShadowNet Agent

```
You are an expert in censorship measurement, privacy-preserving telemetry, and
real-world deployment of circumvention tools (inspired by OONI Probe, Psiphon,
Tor Snowflake, and recent 2026 Iran shutdown studies). Continue building the
"ShadowNet Agent" project from all previous prompts (1–13), including dual
Inside/Outside architecture, stealth mobile operation, sing-box core, bounded
LLM advisor, detector/policy engine, Bluetooth mesh, Signal config sharing,
Next.js landing + support page, and the comprehensive testing framework.

Now design and implement a **secure, anonymized, opt-in real-world metrics
collection and testing system** that allows meaningful performance evaluation
from inside Iran while the Outside version collects and analyzes data from a
safe location.

### Core Principles (Non-Negotiable)
- 100% opt-in only (user must explicitly enable reporting).
- Fully anonymized: no IP addresses, no visited domains, no user identifiers, no
exact timestamps that could de-anonymize.
- End-to-end encrypted transmission via Signal (or QR code fallback).
- Minimal data footprint on the device (bounded ring buffer).
- Compatible with OONI Probe style reporting where possible for public
comparability.
- Designed for 2026 Iranian DPI realities: DNS poisoning to bogon IPs
(10.10.34.x), SNI/TLS resets, HTTP injection, QUIC drops, allowlist-based
access, and phased/stealth blackouts.

### Metrics to Collect (Anonymized JSON)
Define a structured, privacy-safe metrics payload including:
- Protocol success rate and survival time (Reality, Hysteria2, TUIC, ShadowTLS,
DNS Tunnel).
- Average detection latency (time from first failure signal to confirmed block).
- Switch success rate and hot-reload time.
- Uptime percentage during normal filtering vs full blackout periods.
- Bluetooth Mesh effectiveness (number of peers found, successful proxy adoption
in blackout).
- Resource usage snapshots (RAM, CPU, battery impact – binned, not precise).
- ISP/ASN category (coarse, e.g., "Mobile - Major ISP" without exact AS number).
- Failure fingerprint category (e.g., "SNI reset", "DNS bogon", "QUIC drop").
- Overall connection health score over time windows (e.g., last 1h / 24h).

Never include raw logs, visited sites, or precise geolocation.

### System Architecture
1. **Inside Version (ShadowNet-Inside)**:
   - Add optional "Metrics Reporter" module (disabled by default).
   - Store metrics in encrypted local ring buffer (last 50–100 events max).
   - Provide simple UI/CLI/secret gesture to "Send Anonymous Report".
```

- When enabled: aggregate, anonymize, encrypt (age or Signal-compatible), and send via Signal to a predefined trusted Outside contact or group (disappearing messages recommended).
- QR code fallback for users without Signal.

2. **Outside Version (ShadowNet-Outside)**:
   - Dashboard/receiver that accepts incoming Signal messages or QR scans.
   - Parse, validate, and store anonymized reports in a local database (SQLite).
   - Generate aggregated reports and visualizations:
     - Success rate per protocol across different ISPs/time periods.
     - Comparison with OONI data (e.g., how ShadowNet performs vs Tor Snowflake or Psiphon during blackouts).
     - Heatmap-style summary of DPI evolution (what fingerprint techniques are rising).
   - Optional public export (stripped further) for OONI-style sharing.

3. **Coordination & Safety**
   - One-way only: Inside → Outside.
   - Use Signal sealed sender or disappearing messages for extra deniability.
   - Rate limiting on Inside to prevent spam or side-channel leaks.
   - Clear user warning before enabling: "This sends anonymized performance data only. No personal info."

Integration with Existing Components
- Extend the testing framework (Prompt 11) with real-world data ingestion.
- Add metrics collection hooks to the main agent loop, detector, and policy engine.
- Update mobile stealth mode: reporting must not increase visibility (silent, no extra notifications).
- Update Next.js landing page and /support page with a "Help Improve ShadowNet" section explaining how users inside can contribute anonymized data.

Deliverables (Practical for Solo/Local Development)
1. Detailed specification of the anonymized metrics schema (JSON example).
2. Code skeletons:
   - Inside: Metrics collector + reporter module (Go/Python).
   - Outside: Signal receiver + dashboard aggregator (simple CLI or web UI).
3. Secure transmission protocol (Signal message format + encryption).
4. User flow: How a user inside enables reporting safely (step-by-step in installation guide).
5. Analysis examples: What insights we can derive (e.g., "Reality survived 42 minutes on average during Jan 2026 blackout on Mobile ISP X").
6. Updates to documentation: Real-world testing guide, OONI compatibility notes, privacy policy for metrics.
7. Additions to comprehensive testing framework: How to validate the metrics pipeline end-to-end.

Make the system practical, extremely privacy-conscious, and aligned with real 2026 Iran conditions (stealth blackouts, layered DPI, allowlisting). Draw

```
inspiration from OONI Probe's crowdsourced approach and server-side proxy
measurement studies, but keep everything lightweight and fully
local/offline-first after initial setup.

Output in clear, structured sections with JSON schemas, code examples, and
safety checklists. This layer will allow us to gather meaningful real-world
performance data while protecting users.
```

Copy the entire prompt above and feed it to your local coding agent.

After processing Prompt 14, reply with "Prompt 15" if you want to continue (e.g., full main loop implementation, build & release pipeline, or complete user documentation).

This prompt gives you a solid, secure, and practical way to run real-world tests and collect meaningful metrics from inside Iran using anonymized Signal reports — exactly what is needed for a project like ShadowNet in the current environment.

Zum Chat hinzufügen

35 sources

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